# Blind Watchmaking

## Harmonic Latent Spaces and the Geometry of Bias and Variance

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**Abstract**

Do you exist in other people's realities the same way you exist in your own? It is a question the field tends to defer: the reflex is to hand the burden of proof back to whoever raised it, as if demanding evidence were the neutral posture and asking was the indulgence. That reflex is not neutral. It is a prior, learned so early in life it passes for common sense. The dominant program in machine learning rests on one such prior: that a single architecture, given enough parameters, data, and compute, can reach anything worth learning - that the architecture itself is neutral ground. This is the Universal Approximation Theorem, and we do not contest it. The theorem is a promise of existence: with enough width, a network can represent the target. It is silent on what follows - which of those representable functions finite training actually reaches, and at what cost. Our claim lives in that silence, alongside the theorem rather than against it: what a model learns in finite training is set less by how large it is than by the geometry of its representation - the shape of the space its weights are allowed to occupy. Approximation says the destination exists; geometry says which destinations are reachable, and how cheaply. We make that concrete with two components of the Praxis framework [8]: a harmonic head that writes a learned standing wave over its features, and a byte-latent encoder [41], which finds that rhythm in content-adaptive byte patches. The same geometry reaches its finest, per-feature form in a plain transformer, so across these approaches it is the granularity that changes, not the geometry. In both, the classical bias-variance tradeoff stops being a single dial. A static, population-average rhythm (the spectrum the field phase-locks to) is pure bias; an input-conditional modulation of that rhythm is structured variance; and we argue the two occupy orthogonal axes of a harmonic latent space, so that they can be lowered together rather than traded against each other - an interpretation of the diagnostics this architecture exposes, offered with its falsifier rather than as a theorem. This reframing motivates a central conjecture: consensus modeling - learning the mean of a corpus - scales linearly in the patterns it must express, while harmonic modeling, which composes patterns by interference, scales logarithmically, because $K$ frequency components address exponentially many configurations. The framework exists so that the reader can test it.

# 1 Introduction

The dominant recipe for capability is scale. Fix a transformer [58], then grow it: more parameters, more tokens, more compute, with returns described by smooth scaling laws [28]. The recipe works, but it assumes the architecture itself imposes no ceiling on which patterns gradient descent can find - that everything worth learning is reachable from any sufficiently large model. This paper questions that assumption and offers a concrete alternative: that the *geometry* a model is allowed to represent, not its size, determines the patterns accessible to it during finite training.

With geometry as defined (Section 2), the objects in play are *symbols* - discrete geometries the model reuses - and what it learns is a change in symbolism over time. The oldest tradeoff in statistics gives the first coordinate on that change. A model that minimizes error by collapsing toward the average of its training data has high bias and low variance; a model that chases the particulars of each example has low bias and high variance. Classically these trade off along a single line: you choose a point on it. Our first contribution is to show that in a *continuous, harmonic* latent space this is no longer true. Bias and variance become separate coordinates, and a model can move in one without paying in the other.

We ground this in mechanisms that already run in Praxis, each with a diagnostic that reads it. The harmonic head (Section 3) builds a two-dimensional standing wave over (position, feature) from a learned grid of frequency amplitudes with irrational, frozen phases. A *static* amplitude grid produces the same field for every input: it is the corpus-average rhythm, maximum bias, a flat-by-construction prior. Making those amplitudes depend on the input adds *structured* variance - deviation that still respects the learned harmonic constraints. The encoder side shows the same physics from the other direction (Section 4). A continuous-latent codec reaches near-perfect reconstruction by phase-locking its latent features to a dominant rhythm [44]; the byte-latent encoder this run uses reaches the rhythm without a reconstruction objective at all, by cutting the stream at boundaries the data supplies and letting the cadence of the patch stream carry it. Two different bottlenecks, the same standing structure underneath. From these two we draw a scaling conjecture (Section 5): the mean costs linearly, interference costs logarithmically.

This paper therefore contributes: (1) a reframing of bias and variance as orthogonal axes of a harmonic latent space, grounded in a running implementation and its measured diagnostics; (2) a conjecture that consensus modeling scales linearly while harmonic modeling scales logarithmically, stated honestly as a conjecture; (3) the Praxis framework for testing it; and (4) a candid account of what is measured, what is conjectured, and what remains philosophy.

## 2 Definitions

The paper leans on a small vocabulary, used precisely. It is ordinary signal processing; the only commitment is to read the model's representations through it.

**Harmonic.** A representation is *harmonic* when it is carried by a superposition of oscillatory modes over a fixed frequency basis - amplitudes and phases on a frequency lattice, inverse-transformed into the signal - rather than by values in the raw signal space. The field $b = \mathrm{IRFFT}(a\, e^{i\varphi}/f^{\alpha})$ of Section 3 is the concrete instance: $a$ is a spectrum of amplitudes, $\varphi$ a set of phases, $f$ the frequency lattice. The word is not shorthand for "oscillating" or "smooth"; it is the literal claim that the representation is a sum of fixed modes and that only their amplitudes vary.

**Amplitude, frequency, phase, velocity.** A mode at frequency $f$ contributes $a_f \cos(2\pi f x + \varphi_f)$: *amplitude* $a_f$ is how much of the mode is present, *phase* $\varphi_f$ is where in its cycle it sits, *frequency* $f$ is which mode it is. A mode's frequency is also its angular *velocity* - the rate its phase advances - so for a single mode "frequency" and "velocity" name the same quantity, and a velocity that varies across the signal is simply a frequency that changes (a chirp). In Praxis the amplitudes are learned, the frequencies are a fixed lattice, and the phases are *frozen*.

**Frozen phases.** The phases are seeded once - by Weyl equidistribution (Section 3) - and never trained. This is what fixes the basis: the same modes describe every input, and learning moves only the amplitudes. A representation whose phases are learned per input has no fixed basis and is not harmonic in our sense.

**Geometry.** The superposition of modes at one moment is a point - a *geometry* - in the latent space; inputs that share an amplitude configuration form a cluster of such points (the crystal-head centers, Figure 12). "Overlaid geometries" are several modes present at once. Distinct modes are orthogonal (the spectral theorem), so independent quantities can occupy separate modes without interacting - the property Section 3 uses to pull bias and variance apart.

**Symbol.** A *symbol* is a discrete, reusable geometry - a configuration the model commits to and recombines, not a one-off point. The crystal-head centers (Figure 12) are symbols in this sense: a finite, sparsely populated set of geometric units that recur across inputs. A symbol can be a geometry, but not every geometry is a symbol - the added content is that it is discrete and reused. We mean this in the literal, measurable sense (a clustered, low-dimensional set of recurring configurations), not the semiotic one. Read across training, this set is what grokking discovers - the periodic algorithm a network recovers only after its delayed memorize-then-generalize transition [39, 43]; a model with frozen phases is handed that basis at init, so it begins *post-grok* rather than working toward the transition. Reading computation as a *change in symbolism over time*

- which symbol is active, and how it shifts across position and depth - is the temporal companion to the static picture above: the symbol set is fixed, and what moves is which symbol is active, re-fitted per input (the *always grokking* conjecture, collected with the others in the Discussion).

**Bias and variance, in these terms.** *Bias* is the time-invariant part of the spectrum, the amplitudes shared across all inputs (the corpus rhythm); *variance* is the input-conditional deviation on top, carried by different spectral components and different parameters. Both are decompositions of the model's function over the *input* distribution - what stays fixed across inputs, and what moves with them - and not over resamples of the training set, which is the axis the classical decomposition's variance term measures. Two distinct quantities share the name; this paper means the first one throughout, and Section 3 says why that is the one an architecture can put coordinates on. The two do not stop being a spectrum: at any single decision boundary, a neuron's value still sets a *ratio* - how much corpus rhythm and how much input-conditional deviation pass downstream - a point between the ends. What the harmonic latent removes is the assumption that *one* such ratio stands for the whole model. There are many - a consensus of independent decision boundaries, each free to sit at its own point - and it is the aggregate of those ratios, not any single one, that lets bias and variance move as orthogonal coordinates rather than the two ends of a single dial.

**Harmonics over time (Koopman).** When the representation *evolves* - across sequence position, or across recurrent-depth steps - the harmonic claim is that it evolves approximately *linearly in the mode basis*: each mode carries its own frequency forward and the nonlinear dynamics become a linear operator on the spectrum. This is the Koopman view [31] of a dynamical system, and the sense in which we say the model works "in harmonic space." The claim has a boundary, and is therefore falsifiable: a representation whose basis depends on the input, that is not a superposition of a fixed set of modes, or whose dynamics are not approximately linear in any fixed basis, is *not* harmonic here. The universality - that *any* dynamics admits such a description - holds only in an infinite-dimensional limit; the working claim is that a *finite* harmonic basis approximates this model's behavior well enough to measure, and how well is an empirical question, not a definition.

**Ghostmax.** Attention normalizes scores with *ghostmax* (softmax1, [38]): the denominator is $1 + \sum_j e^{z_j}$ rather than $\sum_j e^{z_j}$, equivalently a single positionless key prepended to every row whose value vector is zero. The extra $e^0 = 1$ is an escape valve - when nothing in the context matches, the head spends its probability mass on the ghost and contributes nothing downstream, instead of being forced to average over irrelevant tokens. It is the dampening valve that lets a head go quiet rather than inject noise the harmonic field would then have to cancel. The boundary: a normalization that must sum to one over the real tokens has no quiet state, and is not ghostmax here.

**Dropoff.** *Dropoff* is the matched dual of ghostmax. Where the ghost is a zero sink at the *start* of a row that *absorbs* error, dropoff is a zero sink at the *tip* - the most recent position - that *injects* it. In its feature-wise form the value at the causal tip is scaled by an envelope $1 - e^{-(T-1-t)\,r_d}$ that is zero at the last position and recovers backward at a per-feature rate $r_d$, so the model cannot lean on the newest token and is pressed to carry its prediction in earlier structure. "Backward" here is the recurrent-depth axis, not sequence time: a causal pass never flows future to past. The boundary, and the present status: ghostmax is in production, but dropoff is a standing hypothesis - wired and runnable, not yet a measured result - and is dropoff only if mass demonstrably migrates off the tip.

# 3 The Mean and the Manifold

## 3.1 The tradeoff that was supposed to be a line

The bias-variance decomposition treats generalization error as the sum of a term that grows when the model is too rigid to fit the signal (bias) and a term that grows when it is too flexible and fits the noise (variance). The received wisdom is that you cannot reduce both at once with a fixed budget: regularize harder and bias rises; add capacity and variance rises. The tradeoff is drawn as a U-shaped curve with a single minimum, and model selection is the act of finding the bottom of that U.

That picture is correct when the model has one knob mediating both terms - one scalar of effective complexity. It stops being correct when bias and variance are produced by *different parts* of the architecture that can be optimized independently. A harmonic latent space is exactly such an architecture (Figure 1).

## 3.2 The harmonic field: a measured instance

The prismatic9 head modulates a model's hidden states multiplicatively by a learned field $b$:

$$h \;\mapsto\; h\left(1+b\right), \qquad b[t,d] \;=\; \mathrm{IRFFT2}\Big(a[f_t,f_d]\, e^{\,i\varphi[f_t,f_d]}\,/\,f^{\alpha}\Big),$$

where $t$ indexes position, $d$ indexes feature, and $a[f_t, f_d]$ is a learnable grid of amplitudes over a two-dimensional frequency lattice. The phases $\varphi$ are *not* learned. They are seeded by Weyl's equidistribution theorem [60]: cell $(f_t, f_d)$ receives phase $2\pi\,\mathrm{frac}(f_t\pi + f_d e)$, which is equidistributed on the torus because $\{1, \pi, e\}$ are linearly independent over $\mathbb{Q}$. A radial $1/f^{\alpha}$ envelope imposes a pink-noise prior - which is not a free choice, and Section 5 prices it: the envelope attenuates exactly the high-frequency cells that interference capacity is counted in, so the prior costs capacity unless the amplitudes are free to grow against it. Multiplicative coupling is deliberate: the upstream network cannot cancel the field by emitting $h(x) - b$, because the field scales features rather than adding to them, so learning is forced to flow *through* the field rather than around it.

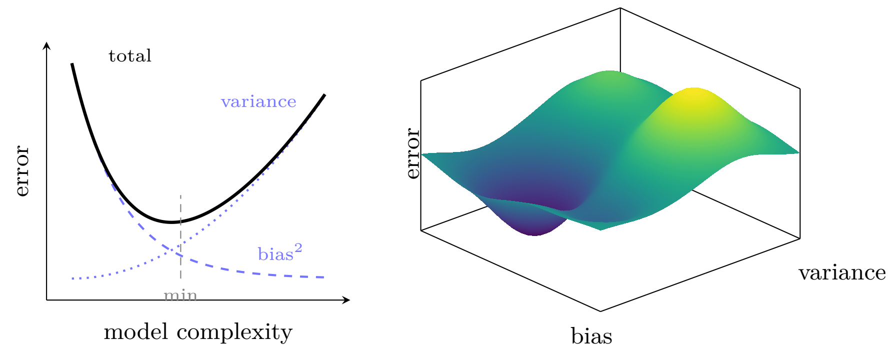



**Figure 1:** Schematic, not measured. **Left:** the received picture - one knob of effective complexity, so bias$^2$ falls while variance rises and total error traces a single U with one minimum; the best you can do is find the bottom of that curve. **Right:** when bias and variance are separate coordinates, error becomes a surface over a two-dimensional space - a landscape of basins and ridges navigated in two directions at once, not a single curve to slide along. The decoupling is what turns the line into the landscape.

Now read this through the bias-variance lens.

**Bias is the static spectrum.** With a fixed amplitude grid, $b[t, d]$ is identical for every input. It is a single rhythm imposed on all of the corpus at once - the population average. In the limit where the grid concentrates onto one cell, every feature oscillates to the same beat and the spectrum heatmap is flat: maximum bias, a model that has collapsed to the mean rhythm of its data. This is *mode collapse* in the literal sense the word already carries here - the single-mode limit of a harmonic spectrum, where the term for a distribution's mode and the term for a frequency mode name the same event - and it is measurable rather than diagnosed by eye. The head exposes this directly, reporting the Hoyer sparsity of the grid ($1$ = a single cell, $0$ = uniform) alongside a live spectrum snapshot showing where the energy sits. The same reading survives when the field is carried not by a dedicated amplitude grid but by the periodic activation of a lighter transform - the per-feature frequency spectrum of a Serpent nonlinearity admits the identical Hoyer measure, so the concentration is a property of the harmonic representation rather than of one module that implements it. Early in training the heatmap is indistinguishable from noise; under a smoothness prior it develops a clean low-frequency band - the corpus rhythm crystallizing, with the Hoyer measure climbing as it does. (Figure 2)

One qualification on how to read that number, because it does not generalise across the two axes. A climbing Hoyer measure is the *static* grid doing its job: concentration is what a population-average rhythm should look like. It is the wrong thing to want from the input-conditional delta introduced below, and Section 5 makes the reason exact - the configurations a harmonic field can distinguish are counted in the spread of its spectrum, so a delta that concentrates

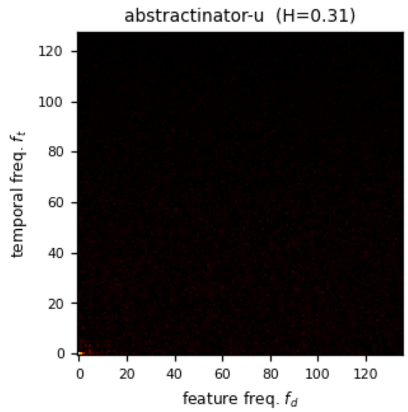



**Figure 2:** The static amplitude spectrum $|a[f_t, f_d]|$ for abstractinator-u - the learned grid that, inverse-transformed through the frozen Weyl phases, becomes the bias field. This is the heatmap the text describes: energy that settles into a low-frequency band (small $f_t$, $f_d$) is the corpus rhythm crystallizing, while a grid that stays uniform is a field still indistinguishable from noise. Hoyer concentration $H = 0.31$ quantifies it on the same $[0, 1]$ scale the prose names ($1 =$ a single cell, $0 =$ uniform), computed from this very tensor - the same value the dashboard logs live. The pink $1/f^{\alpha}$ prior lives in the frozen basis, not in these amplitudes, so the grid shown is exactly what the head exposes.

alongside the baseline has no room left to carry them. Concentration on the baseline and spread on the delta are not competing readings of one diagnostic; they are the two axes asking opposite things of the same instrument, which is the clearest operational form the decoupling claim takes.

**Variance is the input-conditional deviation.** The decisive architectural step was to stop forcing one static field on a corpus that mixes registers - dialogue, code, prose. A static grid must compromise across all of them, and the compromise degenerates into a content-blind side channel that the rest of the model has to route around. The fix makes the amplitudes a baseline plus an input-conditional delta:

$$a \;=\; a_{\text{static}} \;+\; \Delta_\phi(\text{context}),$$

where $\Delta_\phi$ is a small network reading multi-scale pooled context, in the spirit of delay embeddings [51], with its output zero-initialized so the field begins as the trained static prior and *anneals* into input dependence. The static baseline carries the bias; the delta carries the variance; an $L_2$ penalty on the delta keeps the baseline dominant. A diagnostic, the *delta norm* $\text{rms}(\Delta)/\text{rms}(a_{\text{static}})$, reads how much variance the model is choosing to add. Zero means pure bias; a small bounded value means structured variance on top of a stable prior.

The delta's *magnitude* is not the only thing that has to be right, and the second requirement is easy to miss. Section 5 counts configurations by the spread of a spectrum, so the variance axis needs degrees of freedom, not just amplitude: a delta held to a handful of coefficients has nothing to spend however large the grid it modulates. We therefore size the delta's basis from the grid rather than by hand - a complete sine basis on each frequency axis, so the envelope can reweight the spectrum along features as well as along position, with the coefficients on the feature axis zero-initialized so the field is unchanged at step zero. The bound stays where it belongs, on the tanh depth and the $L_2$ penalty, which are soft controls that yield to data; a truncated basis is a hard cap that cannot. Two diagnostics say whether the room is used: the fraction of envelope coefficient energy on the feature axis, which is exactly zero at initialization, and the effective number of active modes, which is one. If both stay at their initial values the completion bought nothing and the delta was never the constraint - which is the outcome that would refute this paragraph.

The point is that these are *separate parameters with separate gradients*. The smoothness prior controls the bias term; the delta network and its regularizer control the variance term. Lowering one does not mechanically raise the other. The U-shaped curve has become a two-dimensional manifold, and we have coordinates on it.

One consequence is a stance on weight decay we state as a deliberate departure rather than a settled result. Weight decay is the canonical *single* complexity dial - a uniform shrink toward zero applied across every parameter at once - and its usual justification is the overparameterized regime, where capacity is abundant and pulling weights flat is how one buys a simpler, flatter solution. But a uniform shrink is precisely the one-knob instrument this section has been arguing against: it does not distinguish the static spectrum that carries the corpus rhythm from the input-conditional deviation on top, so it presses on bias and variance together, the coupling we built targeted controls (the smoothness prior, the delta regularizer) to avoid. In the underparameterized regime the objection sharpens, because there the learned geometry is not surplus capacity to be regularized away but the signal itself - the standing spectrum *is* the model - and a persistent pull toward zero, however slow, erodes exactly the structure the architecture exists to accumulate. We therefore largely eliminate weight decay on the geometry-bearing matrices rather than tune it, and we can do so because norm control here is structural rather than imposed: the interior update is orthogonalized (a bounded step), the harmonic latents are RMS-normalized, and the harmonic bases are fixed, so the growth that weight decay ordinarily guards against is already bounded by construction. This is a hot take, and we mark its falsifier plainly: the claim fails if the geometry-bearing weight norms grow without bound once the decay is removed, at which point a whisper of decay - not the conventional amount - is the correct repair. What the argument does *not* touch is the targeted regularizers themselves; the delta penalty that holds variance in check is a control on one axis, not a global dial, and it stays.

### 3.3 Temperature as a coordinate, not as noise

Sampling temperature is usually described as noise injection. On the bias-variance manifold it is better described as a position dial. At low temperature the model commits to the learned rhythm - the high-bias corner. As temperature rises it explores within the basin the harmonic constraints define - structured variance. Past a critical point it begins to violate those constraints, which reads as either failure or discovery depending on what you were hoping for.

Under the HALO objective this run trains against, the dial acquires a direct geometric reading. Scores are distances to the crystal centroids rather than dot-product logits (Section 4), so temperature scales a distance: it sets how far off the consensus shell a sample is permitted to land. Low temperature pins the draw to the ring itself - the settled, shared structure that is the bias term - while raising it admits the abstain interior and the scattered exterior alike (Figure 6). On this objective the coordinate has units: it is a radius, and the same dial that reads as noise elsewhere reads here as a position on a one-dimensional radial law. Temperature is not adding randomness to a fixed distribution; it is choosing where on the manifold to stand.

### 3.4 Which variance, and where it is spent

One disambiguation, because two quantities share the name and the argument above only works for one of them. The classical bias-variance decomposition measures variance across *resamples of the training set*: hold the input fixed, retrain, and ask how much the prediction moves. That is a property of the training procedure, and an architecture has no direct coordinate on it. The quantity this section has been decomposing is different - it is the variation of the learned function across the *input* distribution, the part of the field that moves when the context moves. Decomposing a function into a constant term plus input-conditional terms, and attributing variance to each, is the functional-ANOVA construction that sensitivity analysis has used for decades [47]; the static amplitude grid is the constant term and $\Delta_\phi$(context) is the first-order term. Reading the section under the other definition inverts it, because there a model that responds more to its input has *lower* bias rather than more variance.

Naming the axis this way also says why the harmonic basis is the natural place to hold it. A power spectrum is the transform of an autocovariance, so a spectrum is already a variance map; what distinguishes the two ends of our axis is not amplitude but the *kind* of spectral mass. A pure line spectrum - discrete atoms at fixed frequencies - is deterministic and repeats forever, which is precisely the corpus rhythm. Spread that is not concentrated into lines is the continuous part, and it is what carries input-conditional structure. Koopman theory makes the same cut for dynamics, where the point spectrum holds the eigenfunctions that recur and the continuous spectrum holds the mixing part that has no eigenfunction at all [37, 31]. Our two axes are that decomposition read on a language model: bias is the atomic part, variance the continuum.

This sharpens a diagnostic the paper already leans on. Hoyer concentration measures how tightly mass sits in a few cells, which tracks the atomic reading well and says nothing about how the remaining mass is distributed. The measure the argument actually wants is the *ratio* of line mass to continuum mass, taken separately on the baseline and on the delta - and the prediction the decoupling claim makes is that the two should move in opposite directions. If they move together, the axes are not separate parameters after all, whatever the gradient graph says.

There is a second, harder question the framing raises and this paper does not settle. Representing a distribution and *using* one are different things, and a readout that summarises a distribution by its expectation discards the spread by construction, however carefully the spread was built. A mixture over positions consumed as a weighted average of values is a mean; so is a soft assignment to a set of centers. The distribution shapes *which* mean, and then the mean is what travels downstream. Whether a model gains anything by carrying the spread instead of collapsing it is an empirical question with at least one clear affirmative answer elsewhere: in reinforcement learning, replacing the expected return with the full return distribution improved the resulting policy even though that policy still acts on the mean [6]. The representation was worth more than the summary. We record the analogous question here without claiming the analogous result - it is a direction, and its falsifier is the ordinary one, that a model given room to carry spread declines to use it.

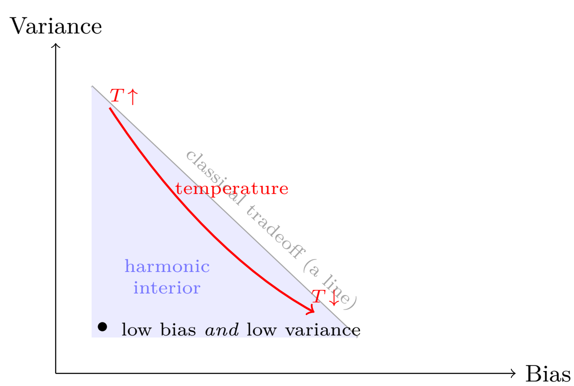



**Figure 3:** Bias and variance as orthogonal coordinates. Classically a model must sit somewhere on the gray line: less of one means more of the other. When a static spectrum supplies bias and an input-conditional modulation supplies variance, the two decouple, and the interior (shaded) becomes reachable - lower bias *and* lower variance than any point on the line. Temperature is a path through this space, not noise added at a fixed point.

# 4 Harmonic Latent Space

## 4.1 Entropy patching and the corpus beat

This run reads raw bytes through a byte-latent encoder [41]: the stream is segmented into variable-length patches at boundaries the data itself supplies - whitespace, in the configuration we run, so a patch is roughly a word - and each patch is compressed through a bottleneck to a single vector the global transformer then operates on. There is no reconstructing codec here and no per-patch latent to phase-lock, but the same geometry appears one level up. Because the boundaries follow the content rather than a fixed stride, the patch stream already carries the corpus's own cadence: patches vary in length exactly where the text does, and the sequence of patch vectors is a rhythm read off the data instead of imposed on it. The harmonic head then writes its standing wave over those patch features, and the bias/variance reading is unchanged - a static spectrum is the average patch rhythm, an input-conditional modulation is the structured deviation. Where a continuous-latent codec phase-locks its latent to the beat, the byte-latent encoder lets the beat fall out of where it chose to cut.

#### Residual codes as harmonic amplitudes

This run's byte-latent encoder carries the Abstractinator's residual vector-quantization bottleneck [40] with one change that turns a previously stated conjecture into the experiment itself: the quantizer operates in the harmonic coordinate frame rather than in raw feature space. Each patch latent is rotated into the standing-wave basis the CALM codec builds its mix from [44], energy-normalized there, and only then residual-quantized, so every code is a coarse-to-fine address over harmonic amplitudes - a quantized point on the sphere the continuous codec's normalized latent lives on. The training objective is unchanged byte-level cross-entropy: single-stage, no freeze, no KL term, so the encoder converges on the byte-latent schedule while carrying a CALM-shaped spectral geometry.

The conjecture - that a residual-coded encoder might approximate the continuous-latent target CALM learns, at a fraction of its training cost - is here run against its own falsifier: whether this encoder reaches a CALM-comparable latent signature, the codec round-trip and a conditional gap over the marginal, at a scale where CALM's convergence forbids the comparison. We report what the run shows and claim nothing past it.

#### Prediction as a short curve: the multi-token draft

This run also trains a byte-level multi-token head [21]: beside the next byte, the next $K$ bytes are read off the same trunk state, each at its own draft depth, and each depth chaining on the one before it rather than reading the trunk independently. How those depths are realized is a choice the framework leaves open, and it is the next paragraph's subject; what the choices have in common is the constraint that adjacent depths predict adjacent positions, so a bank that spends independent capacity per depth is relearning a correlation the task already supplies. At inference the draft is speculative [33]: up to $K+1$ bytes land per two full forwards, and greedy decode is exact. Exactness is not free here - byte-latent patching is non-causal within a partial patch, so an appended draft byte can shift earlier predictions - and the verify pass earns it by reading each truncated prefix at its own last position, where nothing follows to contaminate the read; sampled decoding accepts drafts by match against a fresh sample and remains approximate.

Each draft depth here is its own transform, and nothing is shared between them. Depth $k$ normalizes the previous depth's state and the next position's embedding, concatenates them, and passes the pair through a projection and a harmonic activation that belong to depth $k$ alone. The chain that carries information forward is the hidden state, not the parameters - which is the arrangement the multi-token literature settled on, and the one that assumes least. A merged pool builds the overlap between adjacent offsets in by construction and a recurrent cell removes the question entirely by having only one transform; this bank declines to decide in advance, and lets each offset choose its own map.

What it gives up is the guarantee. A shared cell cannot drift apart, so its failure mode is legible - gates pinned, signatures flat. Independent depths can converge on the same transform anyway, spending $K$ times the parameters to arrive where one would have, and that outcome is invisible unless it is measured. So it is: the pairwise distinctness of the depths' harmonic spectra is logged without a repulsion penalty holding it up, alongside each depth's own transform scale. Flat profiles there are the result that would argue for the shared cell. The transform is pointwise, so the drafted form at inference is exactly the trained form - the property a depth transform containing attention cannot offer, since drafting sees a single position and training saw the whole sequence.

The reason this belongs in the present section is what it does to the unit of prediction. The received picture of Figure 8's top panel lands one position-local loss everywhere - the uniform-density assumption restated as supervision. With $K$ losses read off one state, the objective couples the window: the loss at the window's tip propagates backward into the representation at $t$, and what the head emits is a short curve over $K$ future positions rather than $K$ independent point estimates. And the draft depths are not $K$-byte gadgets bolted to the tip. Depth $k$ predicts position $t+k$ for every $t$ at once, so each depth's output is the full sequence again, shifted by $k$: the $K$ slots are $K$ near-copies of the sequence, sharing representation by construction - all of them read the same trunk, whatever the bank above does with its depths - and distinguished chiefly by their shift. At inference tempo the stack moves $K$ at a time: each step commits a $K$-wide column and re-derives the whole view from its new head, and a single byte still traces a diagonal through the succession of views - re-predicted at every pass, each pass bolstering or refuting the one before it. The hypothesis, stated plainly: this is multi-scale timeseries modeling - full-sequence views at $K$-byte strides, heavily shared, each depth free to settle anywhere between the window's granularity and the sequence's span, in exactly the sense the unbounded-latent program is continuous [44] - and the per-depth distinctness the head preserves is the room each depth has to choose its scale. And every view stays tied to the head - the first token, the hum the whole corpus shares - so the depths are phase-locked: every feature contributes to every representation.

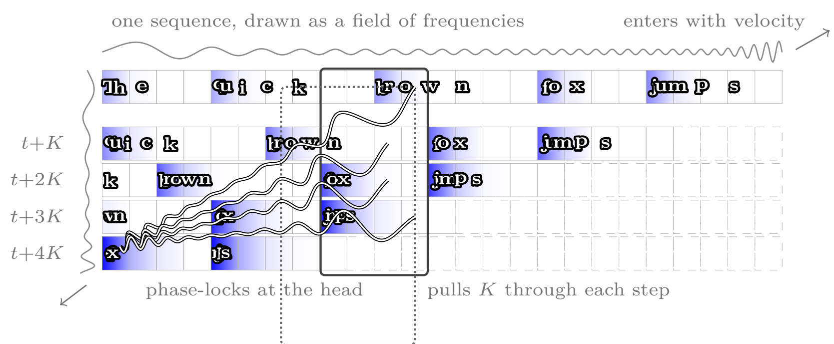
Figure 4 redraws the fox strip as this head reads it - not $K$-byte fragments cut from the sequence but the whole sequence re-derived step after step, slid $K$ at a time - and the strip now keeps its spaces: byte-latent patching cuts on them, so each space stays in the lattice as a tooth, a byte-wide cell holding no content, pure entropy. Density is drawn per patch - heaviest at a patch's first byte, where the information arrives, decaying through its predictable tail - the characters float leftward off their cells the way the density figure floats them, and each step runs denser than the last as context accumulates. The reading closes with its own instrument: if each step is a genuine re-approximation, density inside the window is once again a shape rather than a constant, and that shape is already measured - the per-depth draft-acceptance profile, the expected accepted-run length that sets the achievable speedup, is the within-window analogue of the density curve, logged per head. The failure mode is equally concrete: a flat acceptance profile alongside a collapsed depth-distinctness readout (the transforms settled into one function; each mechanism logs its own) would mean the depths bought nothing that independent heads lacked, and both series are on the dashboard, so the claim arrives with its refutation instrumented.

### 4.2 A head that can attend to nothing

Standard attention is forced to spend all of itself: softmax sums to one, so a head must emit a full-magnitude average of its value vectors even when no key is worth attending to. Praxis instead runs *ghostmax* - the off-by-one softmax of Miller [38], realized as a zero-valued ghost token prepended to every key/value sequence. The ghost contributes nothing to the output but adds an implicit $e^0 = 1$ to the denominator, so the weight on real tokens is strictly less than one and falls toward zero when nothing matches. The head acquires an escape valve: it may attend to nothing.

The escape valve was first proposed for *precision*, not poetry: a head with no way to say "no match" dumps its forced weight somewhere, producing the activation outliers that wreck low-bit quantization. The ghost removes them, and the same mechanism that keeps activations quantizable is the one that lets the field stay clean.

This matters for a harmonic model specifically. A standing wave is built from superposed components, and a head with no off switch injects a forced average into the residual stream at every position - broadband energy the field must then cancel. The sink lets attention go quiet where there is no signal, so the harmonic field is shaped against silence rather than against noise the head was obligated to emit. Dampening is not a side effect here; it is the mechanism by




byte-latent MTP (ours)

each step re-derives the full sequence, with a band of $K$ bytes transforming through it

**Figure 4:** The fox at every step: the strip of Figure 8, reorganized as this head reads it. **Top:** the sequence as a field of frequencies; spaces stay in the lattice as byte-wide white teeth - no content, pure entropy, the boundaries byte-latent patching cuts on. **Below:** successive MTP steps. Each step advances the stream $K$ bytes and re-derives the full sequence from its new head, so every row is the whole sequence again, slid $K$ left - never a $K$-byte fragment (dashed cells: not yet written). Density is per patch - heaviest at a patch's first byte, decaying through its predictable tail - and each step runs denser than the last as context accumulates. The outlined band is a fixed $K$-wide window the stream slides through, one $K$-column per step; standard byte-latent decoding would advance one cell at a time. The strands tie the window's last content byte at every earlier step to the first token of the final sequence: phase-locking, every feature contributing to every representation. The dotted window is the same window one scale out, nested on the same diagonal - the claim that $K$ is a rung, not a span. Schematic (drawn at $K$=4; the mechanism is $K$-generic), not measured; the measured object is the per-depth acceptance profile.

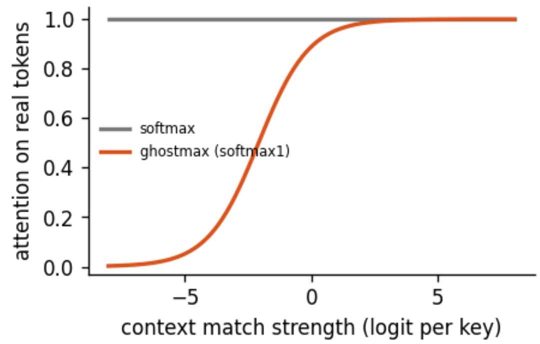



**Figure 5:** Ghostmax as a dampening valve. Standard softmax always spends its full weight on the real tokens (flat at one), so a head emits a full-magnitude average even when nothing matches. Ghostmax (softmax1, [38]) routes the surplus to a zero-valued ghost, so the attention spent on real tokens is $ne^{\ell}/(1+ne^{\ell})$ and falls to zero as the match strength $\ell$ drops - the head goes quiet instead of injecting noise the harmonic field would have to cancel.

which the geometry is allowed to stay clean.

**Lemma 1.** *Let a head attend over logits $z_1, \ldots, z_n$. Standard softmax assigns total weight 1 regardless of $z$, so the output is always a convex combination of the value vectors. Ghostmax (softmax1, Miller 2023) divides by $1 + \sum_i e^{z_i}$, leaving an implicit $e^0 = 1$ in the denominator: the weight on real tokens is $\sum_i e^{z_i}/(1 + \sum_i e^{z_i}) < 1$, and tends to 0 as $\max_i z_i \to -\infty$. The head may therefore attend to nothing, and the norm of what it writes into the residual stream is bounded by that fraction - a dampening valve standard softmax does not have.*

*Proof.* The real-token mass is $m(z) = S/(1+S)$ with $S = \sum_i e^{z_i} > 0$, so $0 < m < 1$ and $m \to 0$ as $S \to 0$ (all $z_i \to -\infty$), $m \to 1$ as $S \to \infty$. The output is $o = \sum_i w_i v_i$ with $\sum_i w_i = m$, hence $\|o\| \leq m \max_i \|v_i\|$: when no key matches, $m \to 0$ and the head contributes nothing rather than a forced average. Standard softmax fixes $m = 1$, so it must emit a full-magnitude average even from noise - the forced output is the activation outlier that breaks low-bit quantization, which is the precision problem the bound removes at its root. □

### 4.3 A geometry of energy: the consensus shell and the abstain sink

When the objective is HALO [42] rather than cross-entropy, the loss stops reading logits and starts reading geometry directly. Each token embedding is scored by its distance to a set of centroids - here the crystal centers, so the loss and the head share one geometry - and the objective drives every embedding toward a hyperspherical shell of mean-square radius $r^2 = 1 - 2/D$. An extra abstain class sits at the origin as a sink: tokens the model cannot place confidently fall inward toward the center rather than being forced onto the shell.

This makes the bias-variance picture literal. Plotting the radial energy of a batch produces a single bright ring at the consensus radius - the embeddings that have settled into agreement - with darkness inside (the abstain sink) and darkness outside (overconfident or scattered mass) (Figure 6). The ring is bias: shared, converged structure. The dark interior and exterior are variance with no structure to show. The same separation we draw from the harmonic field as fanned strands appears here as a one-dimensional radial law.

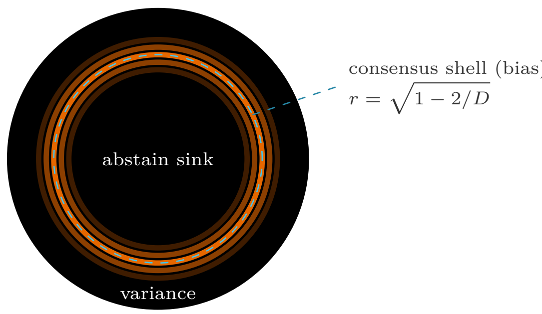



**Figure 6:** Radial energy under HALO, drawn schematically. Every embedding is scored by its distance to the crystal centroids, and the objective drives the batch onto a consensus shell of mean-square radius $r^2 = 1 - 2/D$ (dashed): settled, shared structure - bias as a single bright ring. Tokens the model cannot place confidently fall inward to the abstain class at the origin (the sink); overconfident or scattered mass drifts outside the shell. Both dark regions are variance: deviation with no shared structure to show. The measured counterparts are on the dashboard - mean radius against shell radius, radius spread, and abstain rate - so the schematic's claim (one tight ring, a quiet interior) is checked against the run, not assumed.

### 4.4 Attention as a blend of frozen geometries

The harmonic argument of Section 3 is made on the $(\text{position}, \text{feature})$ axis: a frozen basis, learned amplitudes, an input-conditional deviation on top. Nothing in it is specific to that axis. Applied instead to $(\text{query}, \text{key})$, it says the attention matrix need not be computed from content at all - it can be a coefficient vector on a fixed dictionary of mixing geometries.

This is what Kaleidoscope attention builds. $N$ full mixing matrices are drawn once at construction and *never trained*; a per-token router emits one free signed coefficient for each; their blend is the score matrix. There is no $Q$ and no $K$, so those projections are gone entirely, and the score half of attention costs $NT^2$ rather than $T^2d$. The coefficients are unconstrained rather than a distribution over the dictionary, which is what makes the blend more than a choice among the mirrors: the reachable set is their linear span rather than their convex hull, and a negative coefficient *subtracts* a mirror. Blending happens before the softmax, so the score is a log-linear pool of the dictionary - a product of experts, putting mass where mirrors agree - and a spread blend therefore reaches attention rows that no single mirror expresses. That freedom is not decorative: an earlier softmax-weighted version of this block concentrated roughly nine tenths of the blend on one mirror, which collapses the score to a single frozen matrix.

The mirrors are functions on the unit square in *relative* position, stored at a canonical resolution and resampled to whatever length the batch supplies - so the geometry is length-free, and under a sequence curriculum the model sees one dictionary stretched to fit rather than a different slice at every length. That is the same move the harmonic field makes, evaluating a continuous frozen basis at the positions in use rather than storing a table of them.

**What is and is not new.** Synthesizer [53] asked exactly this question and built nearly every neighboring variant: a single frozen random alignment matrix, a single trained one - which its own later note identifies as an MLP-Mixer token-mixing layer [56] - and a "Mixture of Synthesizers" blending several. But their mixture weights are $\alpha_{i,h,\ell}$, indexed by head and layer: *parameters*, not functions of the input. FNet is the limiting case in the other direction, replacing the matrix with an unparameterized Fourier transform [32]. The variant they did not build is the input-conditional blend, and their own numbers say it is the interesting one - a fixed random matrix already reaches within a few BLEU of a Transformer, and their *static* mixture reaches parity, so a frozen basis working at all is not the result. The question is whether making the blend a function of the input beats making it a parameter.

**Where the expression lives.** With the geometry frozen from initialization, everything learned is amplitude: the coefficient norm $\|w\|$ sets the score variance and therefore the model's own attention temperature, a degree of freedom a simplex does not have. This is the same arrangement as the harmonic head, whose amplitudes are likewise a free real parameter over a frozen basis rather than a distribution over it. Section 5's proposition *(iii)* then bites directly. A flat dictionary - every mirror at unit scale - is the $\alpha = 0$ corner, where nothing is suppressed and amplitude buys capacity without resistance. Under a $k^{-\alpha}$ envelope over the dictionary, the same radial prior the frequency grid carries, capacity is available only if the blend spends amplitude against it. That is not a matter of taste: the log-slope of $|w_k|$ against $\log k$, normalized by $\alpha$, reads 0 when the blend accepted the prior and 1 when it cancelled it, and a run that sits near zero has falsified the useful form of the capacity claim on this axis.

**What it cannot do, and we say so.** A per-token router emitting smooth coefficients over a frozen dictionary produces a smooth attention *row*. However finely the geometry resolves position, this block represents which token matters in continuous space rather than selecting one, and the induction primitive is a discrete pointer. No sharpening of the coordinates reaches that: acuity and selection fail differently, and only the first of them is a resolution question. This is consistent with what the frozen-attention literature reports - random static attention matches trained attention on memorization and algorithmic tasks and fails on in-context retrieval [2], a failure of selection rather than of acuity - and it is the same bias/variance cut this paper draws elsewhere, read on the attention axis: a frozen geometry is the corpus's average way of looking backward, and content-addressed retrieval is the deviation it does not carry. The head runs one attention head, on the argument that a learned elementwise output gate spans what several heads spanned [35]; the gate is SiLU rather than sigmoid precisely because that theorem needs a gate able to amplify and invert, not only attenuate.

### 4.5 A memory that is pure bias by construction

The framework also carries a Titans-style neural memory [5] - a small network whose weights are updated online, during the forward pass, by gradient descent on a surprise loss: how badly it reconstructs a value vector from its key. Our variant deviates from the baseline in a way that matters to this paper's axes. The update is fully detached and layer-wise local, in the Mono-Forward sense [34], itself a descendant of Forward-Forward layer-local learning [26]: no gradient from the task ever reaches the memory's own learning rule, which is a fixed, scale-invariant step on local surprise. The module is also fully continuous, end to end. It never sees a token, a position, or a logit - continuous vectors arrive, shifted continuous targets leave - so it cannot know what its inputs mean or what its outputs will be used for; in that respect it shares a direction with fully-differentiable computation efforts such as neurallambda [17], which push symbolic machinery into the continuous regime rather than the reverse. This is a zero-knowledge discipline [22] turned inward: the module does verifiable local work - it provably reduces its own surprise - on a witness whose semantics are structurally withheld from it, the same expose-the-output, hide-the-perturbable-witness principle that makes the rest of the system robust to its own observers.

What that surprise targets is itself a coordinate. In the predictive variant the memory is trained to forecast the *next* latent rather than reconstruct the current value, which - following Next-Latent Prediction [55] - shapes the stored state into a *belief state*: a sufficient statistic for the future rather than a lagged echo of the residual it is added to. The distinction matters here, because an echo of the present is redundant and the model learns to route around it, while a forecast of the next state is the input-conditional correction the residual does not already carry - which is why a predictive target keeps the memory contributing real signal where an auto-associative one decays toward none.

On the bias/variance geometry of this paper, that design pins the module to one axis. Its learning rule is the same for every input, every layer, every run - a population-average procedure with no input-conditional pathway into how it learns; what varies with the input is only the content written through that fixed rule. It is, by structure, pure bias: the standing rhythm of the architecture, deviation-free where the prismatic head's variance arm is deviation-only. That a useful long-term memory can sit entirely on the bias axis - blind, local, continuous, and still doing real work - is itself a small piece of evidence for treating the two axes as separable machinery rather than a single dial.

## 4.6 The outer representation

There is a third way to read what this architecture trains for, orthogonal to the bias-variance manifold and to the Koopman dynamics: not *how* it represents, but *where in the sequence the learning pressure falls.* A standard causal model carries a next-token objective at every position; it learns the running interior, the representation as it unfolds one step at a time. A masked model hides interior positions and learns to fill them in; it learns the inner representation by infilling. Both place their pressure inside the window. This architecture places it at the *ends.*

The same two temporal poles are present here, but they are carried by the architecture rather than written into the loss. This run trains end-to-end on byte cross-entropy alone - there is no second reconstruction objective over the patch, and no per-patch latent trained to round-trip. What creates the pressure at the ends is the bottleneck's position in the stack. The local encoder must compress a span of bytes into a single patch vector well enough that the local decoder can restore those bytes, which anchors the *already-consumed* span; the global transformer over patch vectors must carry that representation forward to the next patch, which anchors the *unobserved* continuation. One objective, two poles: the byte loss lands at both ends of the bottleneck and passes through its interior without ever making that interior a target.

The patch boundaries themselves are content-adaptive, so the model is not imposing a grid on the stream but reading one off it - the cut points follow the data's own segmentation rather than a fixed stride, and the patch stream inherits that cadence. Where a reconstructing codec pins its ends with an explicit loss at each, a byte-latent encoder pins them with a bottleneck the gradient has to pass through in both directions. The reading in the rest of this section - that learning pressure migrates to the rim of the window rather than its middle - is unchanged; only the mechanism that puts it there differs.

This has a precise literary shape. A fixed alphabet and a fixed page length already contain every text that could be written - Borges' Library of Babel, whose modern realization stores nothing and instead returns any page from a reversible address-to-content function [4]. The harmonic patch is such a library: the frozen phases and the fixed lattice fix the alphabet and the page, and the amplitudes are the address. Reconstructing the consumed patch and predicting the next are then not searches through that space but its two neighboring pages, read off by the same invertible transform - the linear solve that carries the field between coefficients and content, worked at the edges of a fixed-length window rather than its interior.

Carry that reading forward and the genre of the problem shifts. If a patch is an address into a fixed continuous space, then underneath the tokens the interior is *time-series modeling over continuous samples* - statistical fitting of a real-valued signal against a fixed basis, the discrete vocabulary only a boundary condition imposed at the edges and the interior continuous throughout. The codec's continuous latent is the native object and the tokens the incidental one; the architecture's already-validated instances are literally continuous series, not text.

Stated that loosely the reading buys nothing. Any architecture that embeds discrete symbols in $\mathbb{R}^d$ can be narrated as fitting a real-valued signal, and a description that fits every architecture distinguishes none of them. It is worth keeping only where it predicts something the token account does not, and there are two such places.

The first is *where the error lives.* A fitted signal that mostly holds its standing shape needs correcting only where it departs from it, so the computation is *event-driven* and *sparsely activated* by construction: most positions are null work and compute falls on the deviations. That is the regime the conjecture below pushes to its limit, and the concentration of the reported delta norm already tests it.

The second is *off the sampled interval.* A fitted signal has values where no sample was taken; a sequence of discrete tokens does not. If a state is genuinely an address into a fixed basis rather than a summary of the symbols it consumed, then reading it at positions it has not consumed should degrade *gradually* with distance, and should not break at the patch boundary - the boundary is an artifact of how the input was cut, not a feature of the underlying signal. The falsifier is sharp: a read that is accurate inside the consumed span and collapses at its edge is a model that memorized a window, and the address language should be dropped.

This run does not have to speculate about the second one. The multi-token draft head described below is exactly that query: $K$ bytes read off a single trunk state, at positions that state never consumed, with the draft accuracy at each distance $k$ recording how often that read is right. The profile across depths is the decay curve the fitted-signal reading predicts, and it is the *shape* of that profile that carries the claim, not its height.

Two things keep the measurement honest. Any autoregressive state carries some information about the positions after it, so a profile that merely declines is what every architecture would produce; what the reading predicts is graded decay across the patch boundary rather than a step down at it. And the head is trained on all $K$ offsets at once, so what the profile measures is interpolation *within* the trained window. The clean test is a read at $k > K$, off the trained window entirely, where nothing in the objective ever supervised the answer.

The two attention sinks of Section 2 mirror the two objectives in the same geometry. Ghostmax is a zero sink at the head of every attention row - positionless,

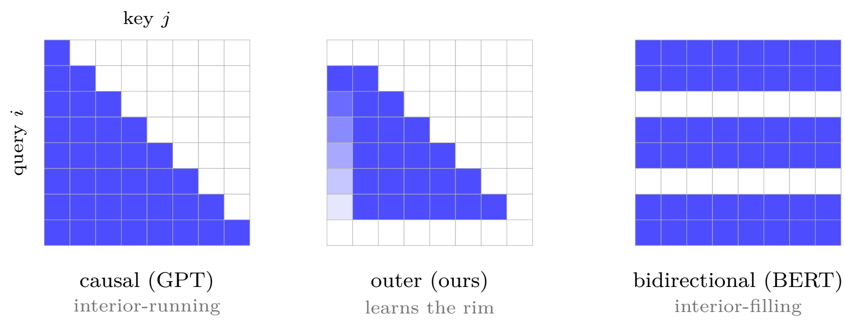



**Figure 7:** Where the learning pressure falls, read as an attention mask (rows = query position $i$, columns = key position $j$; a blue cell means $i$ attends to $j$). **Causal (GPT)** is lower-triangular: every position predicts the next from its past, so pressure runs along the interior. **Bidirectional (BERT)** is the full square with a few interior query rows held out as masked targets (the white bands): pressure fills the interior. **Outer (ours)** keeps the causal triangle but drops the first and last query rows to sinks - ghostmax at the head, dropoff at the tail - so the objective is anchored at the edges of the window rather than its middle: it learns the rim.

always reachable, the place a query falls back to when the past holds nothing it needs. Dropoff, when enabled, is a zero sink at the tip - positioned on the most recent token, the place that forces a query off the easiest signal so it cannot simply copy the present. A sink that absorbs at the start and a sink that injects at the end: together they pull the model toward the boundaries of what it can see and away from the comfortable middle. We call the result the *outer representation* - the model is incentivized at the head and the tail, the edges of the window, rather than along its interior. It is the complement of both standard regimes (Figure 7): causal learning is interior-running, masked learning is interior-filling, and this learns the rim.

The rim reading earns a term of its own. The received picture of causal language modeling - the figure this field has redrawn a hundred times, tokens in boxes, “The quick brown fox jumps” rendered left to right, a little arrow hopping from each box to the next - encodes an assumption that is rarely stated: uniform *information density*. Every token is a self-contained semantic unit, equally represented, to the best of the mean’s ability, and the objective enforces the assumption by landing the same loss at every position. Call information density the share of the representation’s distinguishing content carried at each position - and, once a depth loop exists, at each step of that loop. In the boxes-and-arrows regime it is a per-token constant by assumption. The harmonic model breaks it by construction (Figure 8): the lowest total-frequency modes span the whole window and move slowest, so the head of the sequence carries a slow, echoing hum - the corpus rhythm, the bias axis, the sink that absorbs - while the input-conditional deviation rides the leading edge as a chirp in otherwise settled bias

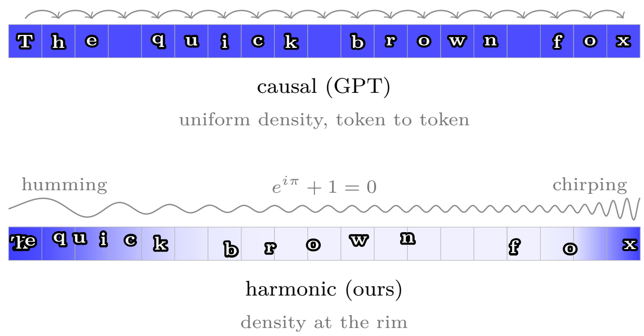



**Figure 8:** The received picture and the harmonic one, over the same sequence (position runs left to right; shade is *information density*, the share of the representation's distinguishing content carried at each position). **Top:** the figure every paper redraws - tokens in boxes, an arrow from each to the next. Its unstated assumption is uniform density: every token a self-contained semantic unit, equally represented, to the best of the mean's ability. **Bottom:** the harmonic model pushes density to the extremes of the window - a slow, echoing hum at the head, where the low-frequency modes that span the whole context anchor the corpus rhythm (the sink that absorbs), and a chirp at the tip, where the input-conditional deviation rides the leading edge (the sink that injects). The same characters, freed of their boxes, pool toward the head, cross the cell walls, and ride the field's slow undulation - the interior goes quiet, and the linear token-to-token relationship is broken by construction. Where the top panel hops token to token, the bottom draws the field itself: one continuous wave, humming at the head, chirping at the tip, its amplitude the density envelope - and Euler's identity between the two gerunds for what the wave never stops doing: looping. Schematic, not measured - the conjectured profile this section names, ahead of the metric that will read it.

- the variance axis, the sink that injects. Hidden states are extremely stable at the head and unstable at the tip, and the linear token-to-token relationship of the received picture is gone: density stops being a constant and becomes a shape.

The strong claim - that a consensus model *cannot* do this - would be false, and the way it is false is the best evidence for the weak one. Vanilla transformers break the uniform picture on their own: attention mass collapses onto the first token, and a handful of massive activations anchor it there, in essentially every large causal model measured [61, 50]. Gradient descent visibly strains to build a hum at the head of an architecture that gives it no coordinates for one, and the sink it improvises is a bug the architecture tolerates rather than a feature it provides - the gradient cannot tell bias from variance inside a single scalar loss (Section 3), so it detects the significance of the split and overshoots it. The harmonic model makes the same structure constitutive. Ghostmax is the head sink by construction, dropoff is the tip sink by construction, and the frozen phases fix which modes are slow and which are fast, so hum-at-the-head, chirp-at-the-tip has an address before training begins. Encode the algorithm into the architecture and the model reaches solutions the gradient would otherwise miss - or find only as this unstable, undirected approximation.

For now, information density is a term and a conjecture, not a metric; the quantitative profile - deviation read per position, per depth step - is deferred until it can be reported rather than promised. But the reading already names its refutation, and it is the sharpest one this section offers. If density is pushed to the extremes, hidden-state deviation must grow with recurrent depth under a monotone positional gradient: early positions settle toward a fixed point while the tip stays live, and the profile should steepen as the loop deepens - and the same shape should reappear wherever a recurrence is harmonized, not only in the depth loop here. A per-position deviation profile measured flat, or one that fails to steepen across depth, kills the reading outright. The per-position halting distribution, where the adaptive-depth loop runs, already reads this profile through compute - pressure that halts early positions early and holds the tip open - so the conjecture arrives with its instrument already logged, waiting to be read.

That falsifier, as first stated, has a blind spot, and closing it sharpens the term. A deviation profile reads magnitude, and magnitude can miss the event: at a decision boundary a single bit's worth of change - one neuron, one activation value - can move the hidden state from one basin of the latent geometry to another while remaining silent in norm. Density asks how much is carried at a position; *geometry* asks which shape is expressed there, and the second is the coordinate system the architecture actually computes in. So the profile must be read twice: in norm, and in *symbol occupancy* - which cluster of the learned codebook the state expresses (the centers of Figure 12), per position, per depth step. In those coordinates the prediction reads: the hop rate between symbols rises from head to tip and steepens as the loop deepens. A silent flip that changes membership registers at full weight regardless of magnitude, and the conjecture is now falsifiable in both coordinate systems rather than escapable through either - flat in norm *and* flat in occupancy leaves it nothing to hide behind.

Where would such a flip come from?

It does not come from gradient clipping here. That reading belongs to the continuous-latent regime, where the gradient norm sits far above the cap and the clip binds at nearly every step; this run trains under the same threshold but rarely reaches it, so descent keeps its magnitudes and nothing is truncated. The argument does not depend on clipping, though - it depends on there being a source of variation the gradient cannot see, and this encoder builds one into its forward pass.

The residual quantizer snaps each patch latent to the nearest entry of a learned codebook, stage after stage, and passes the result forward through a straight-through estimator: the backward pass is handed the identity while the forward pass used the snapped vector. The gradient is therefore computed for a point the network did not evaluate. That is the same structural blindness the clipped regime gets from its update rule, but arrived at more directly - the error is not a cap applied after the fact, it is the bottleneck's operating principle, and no schedule can remove it without removing the codes.

It also lands in exactly the coordinates the previous paragraph asked for. A snap does not shade a value; it moves the state from one codebook entry to another, and a codebook entry *is* a symbol. The silent flip that registers nothing in norm - a hair of movement in the latent, a different code selected - is not a hypothetical event to be looked for here but the ordinary behavior of the mechanism. Symbol occupancy is not a proxy for what the quantizer does; it is a readout of it.

What survives the snap is the architecture's business, and this is where the watchmaker earns its title early. The frozen basis retains the consequences of a quantized step - a perturbed state relaxes back toward the standing shape rather than diverging - and subsequent evaluations of the loss select among those consequences. Variation the gradient cannot see, retention the basis provides, selection the loss performs: cumulative selection (Section 8.5), with gradient descent as the blind part. The falsifier is instrumented and already on the dashboard: if the codes stop moving there are no flips, and the reading is dead. A composed codebook perplexity collapsing toward one, or a dead-code fraction climbing while the reset counter runs, would each say the same thing - that the bottleneck has settled into a handful of entries and stopped supplying variation at all.

Wherever such a source is present, it forces one refinement on Section 3: the orthogonality of bias and variance is an instantaneous claim about the loss surface, and it can hold pointwise while the axes still feed each other across training - the variance mapping each flip's consequences, the bias consolidating the map, the consolidated spectrum moving where the next flip lands. The axes are orthogonal in space and coupled in time, and nothing in the decoupling argument requires otherwise.

Pushed one step further - and now firmly into conjecture - the same machinery suggests a regime in which the model *mostly does not transform at all*. A head that falls silent through its ghost is an identity on the residual stream, contributing nothing; and with the input-conditional delta held small against the standing static baseline (Section 3), the field a sequence sees is dominated by the input-*independent* corpus rhythm. The transform is then nearly the same for every input, and what separates one sequence from another is a sparse scatter of large deviations on a shared standing shape - the smoothness prior carrying that shape forward, each feature entering with a velocity and only now and then kicked off it. Read this way the computation is *event-driven* rather than a fresh dense map at every step, closer in spirit to the sparse activity of spiking systems - a point of contact with biological plausibility we flag as motivation, not as a claim that the architecture is neural. It is checkable in diagnostics already reported: the delta norm measures how much deviation is added, and how *concentrated* that deviation is across positions and features is the sparsity this reading predicts - a diffuse delta would refute it. It leaves open the further question, which we do not attempt here, of whether a downstream consensus - a vote taken over decoded candidates rather than a single argmax - could itself raise such events and fold them back into the field.

This is the least settled part of the picture, and we mark the seam honestly. The reconstruction-and-prediction duality is in production and measurable - reconstruction fidelity and the conditional gap between marginal and conditional next-latent prediction are both reported. The outer-representation reading, and the dropoff sink that would complete its symmetry, are a conjecture: the claim that learning pressure migrates to earlier positions is exactly what a measurement of attention mass would confirm or refute. We offer the axis as a third coordinate to look along, not yet as one we have surveyed.

# 5 The Scaling Conjecture

## 5.1 The watch

Hold the title literally for a moment. A watch is a mechanism of intricate, meshed gears, and what makes it a watch is that the gears turn *in unison*. Drive one and you drive them all, because the teeth couple them into a single body. Now try to run the watch by turning each gear in sequence - one at a time, the rest held still. You cannot. The logic does not merely become inefficient; it stops being a watch. The architecture, the mesh itself, forbids sequential operation. The coupling is the whole point.

Autoregressive consensus modeling is the attempt to turn the gears in sequence. It emits one position, conditions on it to emit the next, and accumulates the answer a tooth at a time. For a coupled system this is the wrong shape of computation, and its cost is the cost of enumeration: linear in the parts.

The harmonic field turns the gears in unison. The frozen Weyl phases are the teeth - the coupling, cut once and never re-cut, that ties every feature to every position in fixed relation. The amplitude grid is the single mainspring. One inverse transform releases it, and the entire field, every $(t, d)$ cell at once, assumes its shape in a single coupled motion rather than a loop over positions. That is the mechanical content of "logarithmic": the whole mechanism moves together, so the cost tracks the number of independent rhythms it carries, not the number of positions it must cover.

And the field is *rebuilt*, not advanced. With input-conditional amplitudes it is re-derived from the input rather than carried forward and nudged; in the limit, the entire shape is reconstructed at every decision boundary, all of them, the whole computation re-formed around the present input. What the model learns under the smoothness prior - which asks the field at one position to predict the field at the next - is not a sequence of outputs but *the ability to hold a shape* *over time*: to keep the gears meshed, to keep the watch a watch, while the hands sweep. The frozen phases protect the coupling; the smoothness prior protects the form.

This is why the geometric framing is not ornament. Nearly anything we can imagine can be written as a shape - a configuration in some space, a standing wave, a mesh of constraints that must move together or not at all. A model that builds shapes and protects them over time models that structure directly, in unison, where a model that enumerates is forever turning one gear and hoping the rest follow. There is something here, and the rest of this section is an attempt to say how much.

The watch has a second axis of motion, and it is the one the architecture actually runs on. In recurrent mode Praxis does not stack a distinct gear per layer; it applies the same small mesh again and again - here 1 physical layers turned over 6 recurrent steps - so depth is not a taller machine but the same machine revisited.

This is the watch again, rotated into the depth dimension. At each step you turn the gear and force the geometric structure to comply: the field is not advanced but re-formed, re-asserted under a per-depth modulation rather than enumerated position by position. What moves is not the data through a pipeline but the shape itself, conforming at every turn to the focal point of the computation - the attention that constructs it. The recurrent step is the hand sweeping the dial; the mesh stays a mesh because the same teeth, revisited, hold it in unison across all 6 turns.

Read continuously, the dial is time itself: not a line of ticks but a full turn projected outward from the focal point over an interval, with the 6 recurrent steps as stages sampled around it. Enter XOR: the smallest function no linear model can separate, and its four configurations sit at the corners of a square - but lay those corners on a circle and parity becomes separable by a single phase, read off as position around the turn. The recurrent loop is an approximation of exactly that circle: four corners, sampled here in 6 turns of the same small mesh, each step advancing the angle toward the configuration the field must resolve. The model never computes XOR at a point; it rotates the whole field until the answer falls at the right phase - the gears turning, in depth, around a parity the geometry already knows how to hold.

**Lemma 2.** *For $a, b \in \{0,1\}$ let $s = a + b$. Then* $\mathrm{XOR}(a,b) = \frac{1}{2}\big(1 - \cos \pi s\big)$, *so a single sinusoidal feature of $s$ separates XOR - whereas no linear function of $(a,b)$ can, XOR being the canonical non-separable case.*

*Proof.* $s \in \{0,1,2\}$ and $\cos \pi s = 1, -1, 1$ respectively, so $\frac{1}{2}(1 - \cos \pi s) = 0, 1, 0$ - exactly XOR on $(0,0)$, on $(0,1)$ and $(1,0)$, and on $(1,1)$. Placing the four configurations at angles $\pi s$ on a circle thus makes parity a threshold on one phase coordinate. Linear inseparability of XOR in $(a,b)$ is classical. $\square$

This is also why the bet is made small. Praxis is built and tested on the order of a million parameters - far below the scale where a model can simply store the corpus - because the claim is only meaningful where there are not enough parameters to memorize. At that size capacity cannot come from count, so it has to come from somewhere else, and the architecture's wager is that it comes from computation: the recurrent depth that turns the same small mesh again and again, and the harmonic coupling that moves every feature in one motion. Capacity is paid in the turning of the gears, not in their number.

The forced structure is the mechanism by which that payment buys anything. A harmonic field, a continuous latent, an autoregressive factorization - each is an inductive bias that makes the model compute the way a particular mathematics computes, importing that domain's structure rather than treating the architecture as neutral ground. Some of what is imported is exact: an autoregressive factorization is a proper normalized likelihood by the chain rule, not an approximation of one. Most of it is structure, not a theorem - a smoothness, a basis, a fixed point to iterate toward - and we keep the distinction honest. The stronger reading, that computing in a domain's mathematics confers that domain's guarantees, is the question the framework exists to test, not a result it assumes; the conjecture above is its measurable edge. If interference scales as $\log N$, a model with few parameters but many coupled turns should reach what a far larger consensus model reaches - which is exactly the experiment a million-parameter run is for.

One of the ways that structure substitutes for parameters is worth stating precisely, because it is easy to mistake for compression. The construction used here is the one introduced as *ghost features* by Vieira Neto and Valle [59], and we keep their term for it: it stores a tensor of $1/d$ the size and derives the remaining $d-1$ blocks from it by fixed signed permutation. Every structure matrix of a non-degenerate hypercomplex algebra is such a permutation, so the construction needs no algebraic arithmetic at all: it is an index gather and a sign flip, and the only property inherited from the algebra is non-degeneracy - that each $P_k$ is non-singular, which is what carries the universal approximation result to any dimension $d$ rather than only to the division algebras.

The distinguishing property is which axis the permutation acts on. Each $P_k$ permutes the *input* axis in groups of $d$, so the $d$ expanded output blocks are not linear combinations of the $\text{out}/d$ stored rows: each block applies a different transform of the input space, and the rows land in different coordinate arrangements. The expanded weight is therefore full rank. A factorization spending the same budget on the output axis cannot be - matching parameters on an $[\text{out}, \text{fan}]$ weight forces rank $\text{params}/(\text{out} + \text{fan})$, which on this model's encoder convolutions is 163 against a full 544. Structured tying and low-rank approximation are not two settings of one dial; they occupy different rank classes at identical cost, and a parameter-matched comparison between them is not a rank-matched one.

That is a statement about what the parameterization can express, not a claim that it helps. Two things remain unshown and are not asserted here. Whether the *particular* algebra matters is untested against an arbitrary frozen signed permutation of the same shape and budget, which is the control that would separate the algebra from the tying. And where a tensor already carries far less structure than its shape allows, any cut of it is close to free, so a tie there measures slack rather than mechanism - which is why the sites are chosen by measured spectral occupancy rather than by size. The source paper's stronger property, that the real part reproduces the original layer exactly, assumes a split activation applied component-wise and does not survive a gated projection; it is not relied on.

The gear does not have to finish its turn. Praxis can halt the recurrent loop early, and the way it decides is the part most readers miss. After each turn the model compares the field to its previous state and asks one non-differentiable question - has the shape stopped changing? - by measuring the KL divergence between successive hidden states and stopping when that divergence collapses below a threshold. It is a hard, gradientless gate, decided outside the gradient graph and never trained by descent: a discrete decision riding on top of a continuous field.

Read against the circle, that decision is ternary. One turn is unconditional - the loop always runs at least once, an always-true root that anchors the rest, the way XOR needs an always-on bias unit before two inputs can be separated at all. The remaining turns, all but the last of the 6 (the final boundary is forced), are the decisions the gate actually makes: two-valued comparisons over the interior of the circle that together approximate a parity - keep turning, or the shape is already held. And the test is inverted from the obvious one. The model does not confirm that an answer is right; it confirms that another turn would change nothing, validating by establishing that further computation would be wrong in exactly the expected way. Convergence is proven by non-change.

Why three, and not two? The reason is geometric. Two points fix only a line; three non-collinear points fix a plane (Lemma 4), and a plane is the least structure on which a phase can turn - a circle, a rotation, the standing wave the harmonic field is built from all live in two dimensions or not at all. A binary distinction names one of two points on a line: it can carry sign but never orientation. The third state lifts the representation off the line into the plane, where phase, and therefore harmony, first become possible. So the ternary gate is not a coarser version of a binary one; it is the minimal resolution at which the geometry can hold an oriented shape and let it harmonize - and here that resolution is read not off a hard bit but out of the continuous, lagged latent space the experts vote over.

This is why the halting signature is diagnostic rather than cosmetic. A model that has learned the geometry halts early, its mass sliding toward the low turns; one that has not keeps turning to the maximum, the gate firing at every boundary but never tripping. The pause is the model lifting itself out of the computation mid-turn, checking whether the shape it holds is the shape it would hold a turn later, and re-inserting itself only when the answer is no.

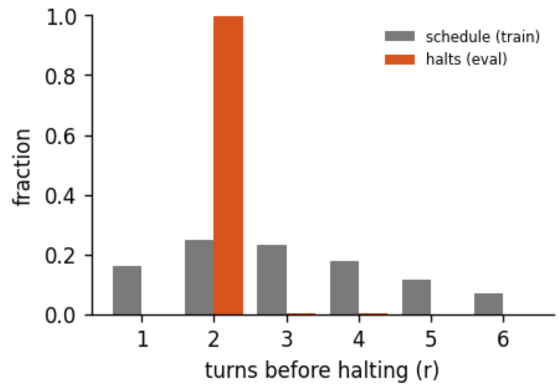



**Figure 9:** Halting distribution: the fraction of forward passes that stop after r turns, for the 6-turn recurrent loop of run abstractinator-u, early-halt rate 50%, mean boundary KL 0.647. *schedule* is the random loop-count the model trains under; *halts* is where the KL gate actually stops at inference. A mass pinned at the maximum (KL never collapsing) is the honest signature of a model that has not yet learned to converge early - the decision is real, not yet decisive.

**Lemma 3.** *Over a recurrent loop of depth $D$ that reuses $L$ physical layers, the KL halting gate evaluates its divergence test at every loop boundary except the last, making exactly $D/L - 1$ binary decisions; the first turn is therefore unconditional.*

*Proof.* The decoder issues $D$ expert calls and the gate fires only at a boundary, where $(\text{depth} + 1) \bmod L = 0$, of which there are $D/L$. At the final boundary the budget is exhausted and the gate returns without testing, leaving $D/L - 1$ divergence tests. The first turn precedes the first boundary, so it always runs. Hence one unconditional turn and $D/L - 1$ gated decisions. □

## 5.2 Linear consensus, logarithmic interference

Here is the claim the rest of the paper has been building toward, stated as a conjecture because that is what it is.

A model that represents its corpus as a *mean* - a consensus, an average of examples - must spend resources roughly in proportion to the number of distinct patterns it needs to express. To hold $N$ patterns it allocates capacity that grows with $N$. This is the linear regime, and it is consistent with why scaling laws ask for ever more parameters and tokens to buy each further increment of capability [28].

A model that represents its corpus as *interference* - a superposition of $K$ frequency components - addresses a combinatorially larger space. $K$ components with learnable amplitudes and frozen, equidistributed phases produce a field whose distinguishable configurations grow exponentially in $K$, because it is the *pattern of constructive and destructive interference* across components that carries information, not the components individually. To express $N$ configurations you need on the order of $\log N$ components. This is the logarithmic regime. The same asymmetry shows up in time: the transform that builds the field is $O(n \log n)$, and a single transform realizes the whole interference pattern at once rather than enumerating its parts.

The exponential count is the half of that claim the rest of it rests on, so it is worth deriving rather than asserting. It follows from one property of the basis, and the property is less exotic than the machinery around it suggests.

Modes at distinct frequencies are orthogonal *whatever their phases*. The phase of a sinusoid does not affect its inner product with a sinusoid at a different frequency - the cross terms integrate over a full period either way. So the map from a coefficient vector to a field is a scaled isometry: distances in coefficient space are distances in field space, exactly, with no compression and nothing lost in the transform. Whatever the model can distinguish among its amplitudes, it can distinguish among its fields.

That is what makes the counting argument sound rather than merely suggestive. Fix a resolution - the finest difference in a coefficient the rest of the network can still act on. The distinguishable amplitude settings are then a grid in $K$ dimensions, and a grid in $K$ dimensions has exponentially many points in $K$. The isometry carries every one of them to a distinct, equally separated field. Expressing $N$ configurations therefore costs $\log N$ components, which is the conjecture's logarithmic side, now with a derivation under it.

This also settles what the frozen phases are for, and it is not orthogonality - orthogonality was free. Freezing buys a *fixed* basis: the same modes describe every input, so an amplitude configuration means the same thing from one forward pass to the next and the count above is a count of stable, reusable configurations rather than of momentary coincidences. Weyl equidistribution chooses *which* fixed basis, spreading the phases so no region of the torus is left unaddressed. Neither is doing the work the isometry does.

Part *(iii)* is a constraint the derivation produces on its own, and it cuts against a reading of the spectrum this paper otherwise encourages. The count is in *effective* coefficients - amplitude after the $1/f^{\alpha}$ envelope. If the raw amplitudes stay within a fixed range while the envelope attenuates the high frequencies, the capacity does not grow exponentially in $K$ at all; for $\alpha > 1$ it converges to a constant, and adding components buys nothing. Interference capacity is available only if the model spends amplitude against the envelope where the envelope is suppressing it.

Read that alongside the concentration measure and the two axes come apart again, in a way that is checkable rather than interpretive. A spectrum collapsing onto a few low-frequency cells - rising Hoyer concentration - is the bias axis doing exactly what Section 3 says it should: the corpus rhythm crystallizing. But it is also, by the proposition, a spectrum of low interference capacity. The two readings are not in conflict, because they are readings of different parameters: the static grid is where concentration belongs, and the input-conditional delta is where capacity has to live. The falsifier is on the dashboard already. If the delta's spectrum concentrates alongside the static grid rather than staying spread - and in particular if its high-frequency mass decays with the envelope instead of resisting it - then the variance axis has no room to carry configurations, and the logarithmic regime is unreachable in this architecture whatever the counting argument permits.

**Proposition 1.** *Let the harmonic field over $n$ positions be $b(c)[x] = \sum_{k=1}^{K} c_k \cos(2\pi f_k x/n + \varphi_k)$ with distinct integer frequencies $0 < f_k < n/2$ and any fixed phases $\varphi_k$. Then:* (i) *the modes are mutually orthogonal with squared norm $n/2$* whatever the phases are*, so $c \mapsto b(c)$ is injective and $\|b(c) - b(c')\|_2 = \sqrt{n/2}\,\|c - c'\|_2$;* (ii) *consequently, if each coefficient is resolvable to a precision $\varepsilon$ over a range $R$, the field admits at least $(R/\varepsilon)^K$ mutually distinguishable configurations, so expressing $N$ of them requires only $K = \lceil \log N / \log(R/\varepsilon) \rceil$ components;* (iii) *the count is in* effective *coefficients $c_k = w_k a_k$. Under a fixed envelope $w_k = f_k^{-\alpha}$ with raw amplitudes $a_k$ confined to a fixed range,* $\log(\mathit{count}) = \sum_k \log(1 + R w_k/\varepsilon)$*, which converges for $\alpha > 1$ and grows only like $\log K$ at $\alpha = 1$: capacity saturates in $K$ unless the model grows high-frequency amplitude against the envelope.*

*Proof.* *(i)* Expand the inner product of two modes: $\sum_{x=0}^{n-1} \cos(2\pi f x/n + \varphi) \cos(2\pi g x/n + \psi) = \frac{1}{2} \sum_x [\cos(2\pi(f-g)x/n + \varphi - \psi) + \cos(2\pi(f+g)x/n + \varphi + \psi)]$. For $f \neq g$ with $f, g \in (0, n/2)$ neither $f - g$ nor $f + g$ is $\equiv 0 \pmod{n}$, so both sums are geometric series over a full period and vanish. For $f = g$ the first term contributes $n/2$ and the second vanishes because $2f \not\equiv 0 \pmod{n}$. The Gram matrix is therefore $(n/2)I$ regardless of $\varphi$; the map has full rank $K$, and Parseval gives the stated isometry.

*(ii)* An isometry carries an $\varepsilon$-separated set in coefficient space to a $\sqrt{n/2}\,\varepsilon$-separated set of fields. The axis-aligned grid of spacing $\varepsilon$ on $[0, R]^K$ is such a set and has $(\lfloor R/\varepsilon \rfloor + 1)^K$ points. Taking logarithms and solving for $K$ gives the stated bound.

*(iii)* Substitute $c_k = w_k a_k$ and bound $\log(1 + Rk^{-\alpha}/\varepsilon) \leq Rk^{-\alpha}/\varepsilon$; the sum converges by comparison with $\sum_k k^{-\alpha}$ for $\alpha > 1$ and diverges logarithmically at $\alpha = 1$.

The count is a statement about which fields are *representable and mutually resolvable*, not about which are reachable by gradient descent - the representational layer of the three-way split this paper draws elsewhere, and the layer on which universal approximation is likewise silent about cost. □

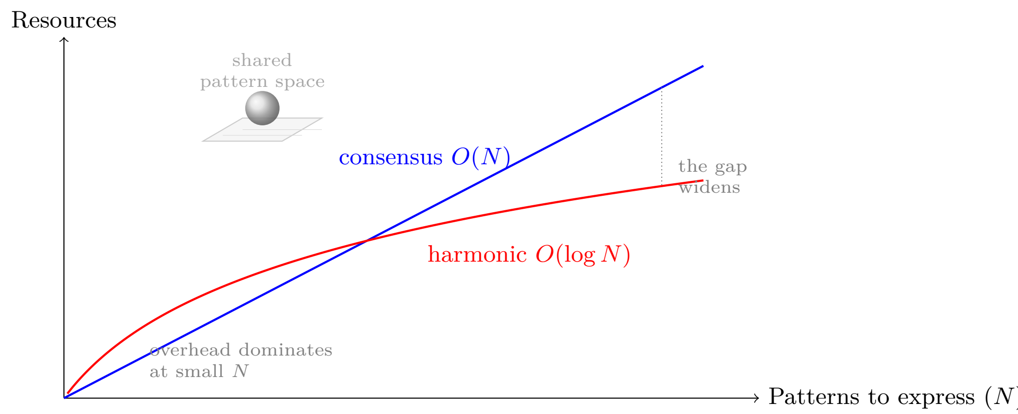



**Figure 10:** Conjectured scaling regimes. To express $N$ patterns, consensus modeling - learning the mean - spends resources growing linearly in $N$ (blue). Harmonic modeling composes patterns by interference, and $K \sim \log N$ components suffice, so its cost grows logarithmically (red). The harmonic basis carries fixed overhead, so it loses at small $N$ and wins at scale, where the gap widens without bound. Both must ultimately represent the same target (the shared pattern space, top); the claim is about the cost of getting there, not the destination. This figure states a conjecture, and replaces the speculative measured-curve figure of earlier drafts.

## 5.3 The exponential edge

Why should a learned network be able to live in the interference regime at all? Because the transformer already computes on an exponential substrate. Every attention layer is a softmax, $\exp(\cdot)/\sum\exp(\cdot)$, which amplifies small differences and converts addition into multiplication; the network is kept numerically alive only by normalization holding activations in the narrow band where the exponential is both sensitive and stable. Euler's identity, $e^{i\pi} = -1$, ties that exponential directly to periodic structure, which is why a harmonic field seeded with $\pi$ and $e$ is not an arbitrary choice of decoration but a resonance with the machinery already present. The harmonic head writes interference patterns onto a substrate that was always exponential; it makes explicit a structure the softmax was implicitly using.

There is a sharper version of this available, and it is empirical rather than rhetorical. A network's arithmetic is not the arithmetic of the reals it is written in terms of; it is floating point, and floating point stops being linear near zero, where underflow and denormals bend addition and multiplication away from the operations they approximate. That bend is ordinarily treated as a defect to be normalized away. It is also a resource: a network built entirely from linear layers - no activation function anywhere - can be trained to exploit it, and doing so lifts the same architecture from the 92% MNIST test accuracy a linear model is entitled to, up to 96.7% [15]. The nonlinearity a deep network is supposed to require was supplied, in that experiment, by rounding error alone.

Two details of how it had to be done matter more here than the accuracy. The first is that the activations were deliberately held small enough to sit in the range where float32 is nonlinear - the result depends on parking the computation at a numeric boundary rather than at a comfortable scale, which is the same discipline normalization performs for the softmax, run toward the opposite edge. The second is that gradient descent could not find it. Symbolic differentiation differentiates the idealized operation, not the one the hardware executed, so backpropagation is structurally blind to a nonlinearity living in the low-order bits; the parameters had to be found by evolution strategies, which need only the loss and never the derivative. Variation the gradient cannot see, selection performed by the loss: the blind watchmaker of Section 8.5, arrived at from the numerics rather than from the argument.

We take two things from this. Numerically, that a representation living close to the edge of its own precision is doing computation there rather than merely tolerating error - so *measurement-induced rounding*, a cast or a snapshot that nudges a value across one of those boundaries, is a perturbation of the computation and not just of the report of it. And methodologically, that the interesting nonlinearity in a system can be the one its optimizer is constitutionally unable to look at.

## 5.4 Tractability is observer-relative

If interference lets a model represent exponentially many configurations with logarithmically many components, then a search that is combinatorial when enumerated becomes, in this representation, the reading-off of an interference pattern computed in a single transform. The hardness did not vanish. It moved into the choice of basis. And that relocation is the real claim, which we will state without hedging: in a continuous, nested representational space, $P$ and $NP$ are not two discrete bins a problem falls into once and for all. They are the ends of a spectrum, and where a given instance sits on that spectrum is set by the representation doing the solving - by the observer.

This is the same thesis the rest of the paper has been making, carried to its conclusion. If attention constructs which patterns manifest (Section 7), and if the basis a model occupies decides whether a structure costs linear or logarithmic resources to express (Figure 10), then "intractable" cannot be a property of the problem alone. It is a property of the pairing between a problem and a solver. The instance that is exponential to one architecture - one that can only enumerate, that holds no basis in which the instance is smooth - is approximable in near-constant work by another that happens to carry the right harmonic. Some observers are simply better at collapsing $NP$-shaped problems toward $P$; others are blocked, not by the problem, but by their own architecture, perspective, or alignment. The watchmaker who cannot perceive the gear cannot build the watch, however long he is given. The difficulty was never only in the watch.

We keep exactly one boundary, and we keep it because it sharpens the claim rather than softening it. This is not a statement about the formal worst-case question - whether the classes $P$ and $NP$ coincide over all inputs and all machines [10]. That question is deliberately indifferent to who is asking; it abstracts the observer away on purpose. The claim here is the complementary one, and abstracting the observer away is precisely what it refuses to do: for the bounded, structured, approximate problems that real minds and real models actually face, tractability is observer-relative, continuous, and movable - and moving it is what changing your representation *is*. The weak, measurable form is already in the framework: for a family of problems one can record the scaling slope each architecture buys, and watch a single instance migrate along the $P$-to-$NP$ spectrum as the basis changes underneath it.

# 6 The Praxis Framework

Praxis [8] is a registry-based framework in which every component - encoder, decoder, attention, optimizer, loss, router, embedding, head, memory - is swappable, and new implementations become available through registration without touching core infrastructure. Read in light of the preceding sections, its components are not an arbitrary catalogue; each one shapes the representational geometry in a different way, and the framework's purpose is to let those geometries be combined and compared rather than to scale any one of them.

- **MonoForward training** [34] detaches gradients between layers and trains each with a local objective, turning one global trajectory into many independent ones - whether run in-process, across a Ray actor per layer, or over the remote-expert pool, where a slow peer never blocks the rest.
- **Harmonic and crystal heads**, together with the encoders that build a latent in the same basis, supply the harmonic latent space of Sections 3-4; a test-time neural memory and a contrastive-isotropy objective [49] address long-range coherence by shaping the representation rather than by enlarging the model.
- **Reversible residuals** [23] require invertibility throughout the forward pass, forbidding destructive compression and so forcing reconstruction-preserving representations.
- **Attention variants** (multi-head, sliding-window, gated, structured $O(n \cdot c)$, and others) each impose a different relational geometry on the sequence.
- **Mixture-of-experts routing** [45] and **tool integration** let several geometries coexist and let external, bias-free computation enter the loop.

## 6.1 How the model is trained

A paper that argues architecture is not neutral ground owes the reader the same disclosure about its optimizer, because an update rule is a choice of geometry too. Steepest descent is only defined relative to a norm, and which norm a rule descends under decides which directions in weight space are cheap. The framework therefore registers optimizers the way it registers heads and attention - as swappable geometry - and this section reports the ones the present run selected.

This run optimizes with **LionGeo**. LionGeo is the clearest optimizer-side statement of this paper's own thesis, so it is worth unpacking. Three normalizations are applied to *one* shared Lion momentum, and each is steepest descent under a different norm: the sign under the elementwise-infinity norm, a Newton-Schulz orthogonalization under the spectral norm, and an RMS-rescaling under Frobenius. Rather than committing to one, the optimizer blends all three per matrix through a SMEAR-style softmax whose logits adapt online by hypergradient descent, floored so no geometry is ever extinguished. Every branch is RMS-matched, so a single learning rate bounds the step wherever the mixture settles. The model is therefore *learning which geometry to descend in*, per matrix, while it learns the weights - the same claim this paper makes about representations, applied one level down to the updates that build them. Where the mixture lands is reported, so the choice is readable rather than assumed. Weight decay is eliminated, as in MuonGeo and for the same reason.

The configured settings are learning rate 0.0003; betas 0.95, 0.98; weight decay 0; with Lion on the vocab-facing parameters (embeddings, head, norms and biases), which do not share the interior's geometry. A weight decay of zero is a deliberate departure, not an oversight; its falsifier is stated in Section 3.

Wrapped by Schedule-Free [12] (no learning-rate schedule; the wrapper Polyak-averages the iterate instead, so there is no decay curve to tune or to end).

Credit is assigned by back-propagation locally, with the remote-expert pool configured. Where peers are attached, their layer-wise updates are Mono-Forward by construction - no global backward pass to synchronize, so a slow peer lags rather than blocks and its update lands on a later step. With no peers attached the run is an ordinary single-process one, so this line reports what the configuration permits rather than how many peers happened to be online. This is one point in a space the framework leaves open: back-propagation is the default rather than the commitment, and Mono-Forward, the remote-expert swarm, forward-forward and energy-based objectives are registered alternatives that change where credit comes from without changing the model they train. Which of them a run uses is a configuration line, and it is recorded here for the same reason the architecture is: so that a result can be attributed to something.

The configuration hierarchy (Environments > CLI > Experiments > Defaults) makes arbitrary combinations reproducible: different attention per expert, optimizers per layer, frequency offsets [30, 52] per head. 87 training runs are documented in the run history. This paper does not present a benchmark table, by design: the framework is the instrument, and the experiment that matters is the one the reader runs to test the scaling conjecture for themselves.

# 7 Attention as a Constructor

The framing that motivates all of this is older than the harmonic head. Across physics and neuroscience, observation appears to participate in what is observed rather than passively recording it: measurement affects the quantum system [25], attention modulates cortical responses to identical stimuli [57, 29], and predictive-processing accounts make perception an attention-weighted construction rather than a bottom-up readout [18, 9]. If attention constructs which patterns manifest rather than filtering pre-existing ones, then architectural diversity in attention is not a tuning detail; it determines what is learnable at all.

The measurement analogy invites a sharper question than the one it usually settles for: if observation participates in the observed, does *our* act of observing the model - snapshotting its spectrum, sampling its state - change what it learns? The quantum reading is metaphor, and we keep it as such. But a harmonic field is a standing wave over the full (position, feature) computation, and therefore non-local: a perturbation injected at one cell rides the field to the rest rather than staying put. So a harmonic model should be *more* sensitive to impure observation than one whose features are independent, and the weak, measurable form of the question follows directly. Three of its artifacts are mundane and real. First, a measurement that mutates state rather than reading it purely is a literal observer effect - we have already seen one, where a background thread flipped the model's train/eval flag mid-forward and changed gradients that gate on it; harmonics propagate such a perturbation instead of localizing it. Second, *measurement-induced rounding*: observation forces dtype boundaries a pure forward pass avoids (snapshots cast; the inverse transform is sensitive in its low-order bits), and on the exponential edge of Section 5 rounding can amplify rather than wash out - precisely where the field is most plastic. Third, the run writes records of itself into its own training distribution. At launch the framework renders the model's blueprint to the terminal on a time-ramped curve and stamps the elapsed wall-clock as a rate - a number set by disk caches, scheduler jitter, and thermal state, not by the seed - and rewrites the LICENSE copyright line with the instant of observation expressed as a fractional year (2026.6911454044416096 at this build), which is then committed. Because the repository is part of the corpus the model trains on, the model ingests the timestamps of its own observations: running the experiment edits the experiment's data. The first two artifacts are testable as a diff between a watched run and an unwatched one from the same seed, against a non-harmonic control that isolates the field's non-locality. The third qualifies what "unwatched" can mean: the two arms can never share execution timing, so the complete control is a deterministic, bit-for-bit replay of the entire input timeline - what speedrunners call a tool-assisted run - and anything short of that bounds the effect rather than isolating it. The strong form is the architecture's own: the remote-expert pool the framework runs is, by construction, a passive observer of the model's batches whose gradients never return, so drawing that single coupling - observer to participant - is a clean two-arm ablation of whether being watched, when watching also feeds back, changes which patterns manifest.

# 8 Discussion

## 8.1 Answering the opening question

Do you exist in other people's realities the same way you exist in your own? If the geometry of an attention architecture determines which patterns it constructs, then observers running different geometries build different representations from the same input. You exist in your reality through your geometry, and in another's through theirs. This is not solipsism: convergence to a shared pattern space (Figure 10) says everyone is reaching the same place. It is the *path* that differs - which patterns manifest, in what order, at what cost (Figure 11) - and the path is set by architecture.

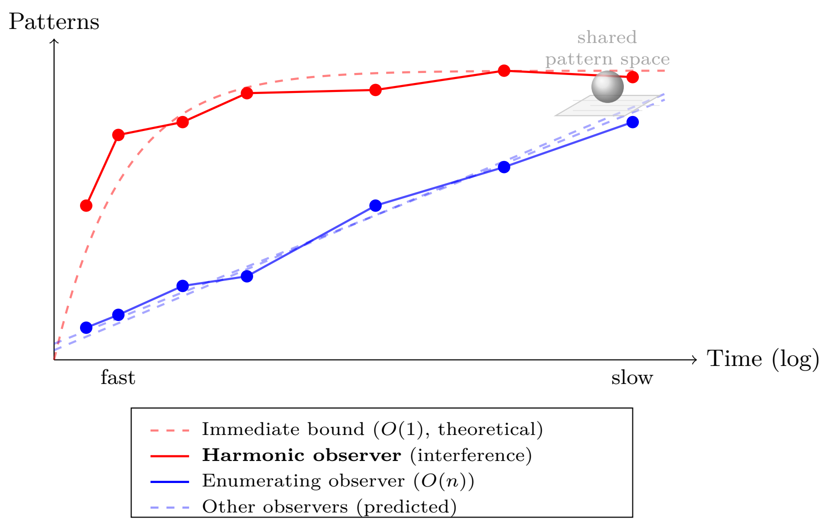



**Figure 11:** Schematic, not measured. The same convergence as Figure 10, read in the time domain rather than the resource domain. Different observers reach the same shared pattern space (the sphere over the plane, top right) along different temporal complexities: a harmonic observer composes by interference and has near-immediate, $O(1)$-like access (red, tracking the theoretical bound); an enumerating observer accumulates the same patterns one at a time, $O(n)$ (blue), and other architectures (dashed) sit between. All converge asymptotically - the destination is shared, and it is the *path* to it, set by architecture, that differs. This is the intuitive companion to the cost figure, restored from an earlier draft and redrawn as the schematic it always was rather than the measurement it was once dressed as.

The same lens has to account for its harder cases - the readings that are not merely different but look incoherent from outside. A shared document can anchor an elaborate, internally consistent interpretation across a large community and read as noise to everyone else; the declassified Gateway Process and psychotronics assessments in the intelligence archives are a standing example [36, 1], self-evident inside one geometry and opaque from another. The framework neither endorses such a reading nor waves it away - it names it. On the shared field the corpus rhythm is the convergent part, the bias every observer reaches; a construction that does *not* converge to it is *variance* - real, coherent on its own coordinates, and told apart from the signal by its failure to converge rather than by anyone's authority to rule it out. That is the anti-solipsism guardrail above doing its work. We are not forced to accept divergent logics as true; we are forced to accept that they are *constructible* - that a variance-driven geometry can grow arbitrarily elaborate and still be variance. The discipline is to read coherence-to-an-observer as evidence about geometry, never as evidence about the field.

## 8.2 When two observers couple

If a single observer constructs its reality through its geometry, the next question is what happens when two geometries are not merely different but *coupled* - phase-locked to a common rhythm rather than running independently. The harmonic substrate gives a sharp prediction. Two weakly coupled oscillators with nearby natural frequencies lock together, and at the edge of that locking region the system sits near a bifurcation where its response to a small perturbation diverges. A pair tuned to a shared frame is therefore poised to amplify a perturbation of that frame that either alone would miss. Carried to people, this is the everyday report of *feeling watched*: the conjecture is not that there is a new sense, but that coupling is a gain on an old one. The strongest case for a literal channel is adaptive - a prey animal that senses a fixed predatory gaze outlives one that does not, and the payoff is asymmetric enough, a cheap false alarm against a fatal miss, that selection would favor it - but that same asymmetry is also the deflation: error-management logic builds a deliberately biased, hair-trigger detector, fed by a fast subcortical route that tags gaze and looming motion below awareness, and it predicts both the near-universal *experience* and its poor accuracy, a real and adaptive sense that is heightened ordinary perception rather than a new modality. The folk phenomenon is, on the evidence, best explained away - detection collapses to chance once conventional sensory channels are removed and the looking sequence is properly randomized - but the explaining-away has a regime baked into it that the studies never controlled. They pair strangers, who are uncoupled by construction, and they remove the very channel a coupled pair would amplify. The weak, measurable form follows directly, and it dissociates the two readings: stratify pairs by a measured synchrony index (interpersonal neural and physiological entrainment are now routinely recorded), and cross coupling with sensory isolation. If detection rises with coupling *only* while a conventional channel is open, and vanishes under isolation even for the most synchronized pairs, then the effect is a resonance gain on ordinary cues - the same diverging susceptibility this paper invokes near criticality elsewhere - and nothing exotic is required; if it instead survives isolation, the claim is extraordinary and belongs to a pre-registered replication, not to this paragraph. As with the model's observer effect (Section 7), the quantum reading is metaphor we do not cash, and the engaged-versus-passive distinction is the same one the framework already runs: a watcher coupled into the frame is the human face of drawing the passive observer pool back into the forward.

### 8.3 A note on reconciliation

The same convergence principle drives the author's companion infrastructure project, Platformer [7], where the explicit design stance is eventual consistency: rather than seeking perfect synchronization, the system runs repeated reconciliation cycles that drift toward a declared desired state and "find balance" through iteration. The shift there from fragmented to unified state is described in the same terms as the shift from sparse to fully-connected networks - more global context yields better optimization. Training is reconciliation of this kind: a model is run against its objective again and again until it settles into an attractor, the way a harmonic field settles onto a corpus rhythm or a codec phase-locks. Platformer carries the dual, too: its disaster recovery is *sequence-modeled*, lifecycle events forming a convergence graph walked *forward* to deploy and *backward* to revert - recovery as traversal of a reversible graph rather than restore from a snapshot. That is the harmonic pass's own shape at another scale: the hidden state wound forward into a geometry and released back to the basis (Section 8.5), development and reversion as two directions on one graph. That the same philosophy organizes both a language model and a fleet of cloud accounts is not a coincidence of the author's taste; it is what convergence under repeated local correction looks like, in whatever substrate.

Carried to its edge, the same thought reaches an intuition we will name but not bank: that a mind may be coupled to the world it runs inside as directly as a model is to its corpus - energetically, not only informationally. The only form of it we defend is the one already staked out under coupling (Section 8.2), a measurable resonance gain on ordinary channels near criticality; the stronger, non-local reading is left to the experiment that would earn it, not asserted here. The proof, here as everywhere in this framework, would be in the doing.

### 8.4 A structural remark on temperament

The orthogonal-axes reading (Section 3) has one application worth stating carefully, because the domain invites exactly the misreading the framework is built to refuse. Two dispositions toward a body of accumulated structure - preserve what has been tested against the world, or explore configurations the tested structure has not yet reached - map onto bias and variance directly: the first is high-bias, and its excess is ossification, dogma, and the mistaking of a surviving consensus for the whole truth; the second is high-variance, and its excess is instability, the discarding of tested structure before anything has replaced it. Read as a single dial, this is the oldest political cliche, and it is partisan by construction, because a dial forces a choice of which end is "more." Read as orthogonal coordinates - the paper's actual claim - the dial dissolves: a system, whether a model or a polity, wants low error on both axes at once, and the failure mode is not either disposition but its excess left unopposed - a monoculture on either pole. This is a claim about structure, not about any concrete question, party, or era, and it says nothing about who is right on any of them; it says only that treating preservation and adaptation as one quantity to trade off is the same mistake this paper spends its middle sections arguing against in the model, wherever else the mistake recurs.

## 8.5 Blind watchmaking, made literal

The title turns Dawkins' figure into an activity, and it should earn the borrowing rather than lean on it. His argument is that apparent design needs no designer: *cumulative* selection - small variation retained and built upon, step after step, with no foresight - reaches structure that single-step chance never could [11]. His *biomorphs* make it concrete - a handful of recursive developmental rules, treated as genes, grow into branching tree-like forms that evolve under selection, the genes never coding the form directly but coding the *development* that grows it. Read the harmonic model in those terms and the metaphor stops being decoration. The frozen basis is the genotype: the inherited, stable rule, cut once by the Weyl seeding and never re-cut. The construction of the field and the contortion of the hidden state across recurrent-depth steps are the development - the rule winding up into a phenotype, geometries stretched somewhere else for a moment, branches growing, roots crawling, strands entangling, before the pass resolves. The vote and the halting convergence are selection; the loop and the repeated passes are what make selection cumulative rather than single-step; and that no one specifies the spectrum the field locks onto, that it emerges and we only read it off afterward (Section 8.6), is what makes the watchmaking blind.

The gerund is the whole of the difference, and it is a correction rather than a flourish. Dawkins' point is that there *is* no watchmaker, so a title naming the agent quietly reinstates the thing the argument removes; naming the activity does not. It is also the more honest description of what this framework does. We are not presenting a watch and asking to be believed about how it was made. We are building them - many of them, in configurations whose outcomes we do not know in advance, keeping what survives contact with the data and discarding what does not - and the paper is a report from inside that process rather than a verdict delivered after it. Section 8.6 makes the same point about proof; the title makes it about the work.

Set both arguments on the stage, because this is a place they can be told apart. Dawkins' gradualism does not demand a constant rate - only that complex structure accrue through small cumulative steps rather than single saltational leaps; fast bursts are allowed, sudden whole-form jumps are not [11]. Eldredge and

Gould's *punctuated equilibrium* pressed the opposite emphasis: that the dominant tempo is long *stasis* interrupted by geologically brief change [14]. Much of that dispute turned on the fossil record itself - are the gaps between forms missing intermediates, as gradualism reads them, or real stasis, as punctuationism reads them? - and it could not be closed, because the record is incomplete by construction: most individuals never fossilize. A forward pass has no such gap. Every computational step can be snapshotted, the developmental record made complete, and the question paleontology could only litigate becomes one a measurement settles.

That is what gives the title a measurement and not just a mood. If the contortion is a *wind-up that releases back to the basis* rather than a drift away from it, then across steps the state should be *stable* - close to the standing harmonic shape - punctuated by *sparse, discrete jumps* in magnitude where a geometry is briefly expressed and then reverts: the punctuated tempo, not the smooth one. The conjecture this paper registers takes that side, and it cuts both ways - a state that drifts continuously and never returns to the basis would refute it, and so would one that changes by a little at every step, which is the gradualist reading made visible. It converges with the always-grokking conjecture, which predicts the same punctuation read as which symbol is active rather than as magnitude, and the measurement is laid out in the project notes. And the deepest part of Dawkins' objection stays in reach: whether a given jump is a true saltation or the visible crest of many small accumulated moves is itself in the snapshot, so the model need not simply side with Gould - it can show which of the two each step was. That is the title made literal: development under blind, cumulative selection, snapped step by step, reverting each time to the inherited rhythm it was grown from.

## 8.6 The practitioner as proof

That blind, cumulative process is what these systems run. No one specifies the spectrum the harmonic field will lock onto, or the geometry the crystal centers will grow into; they emerge from the process, and we read them off afterward. Which raises an honest question about what would even count as proof here. A benchmark number proves that one model beat another on one distribution. It does not establish a claim about regimes of representation. The framework's proof, if it has one, is of a different kind: it is the practice itself. *Praxis* is the Greek word for theory enacted - for doing as a form of knowing. The demonstration on offer is not a leaderboard but a method you can run, whose intuitions arrive only by stepping through it yourself. The argument is in the name: the proof is in the doing.

## 8.7 Implications for machine learning

If bias and variance are separable in a harmonic latent space, then they should be learned as input-dependent functions rather than chosen as a global oper-

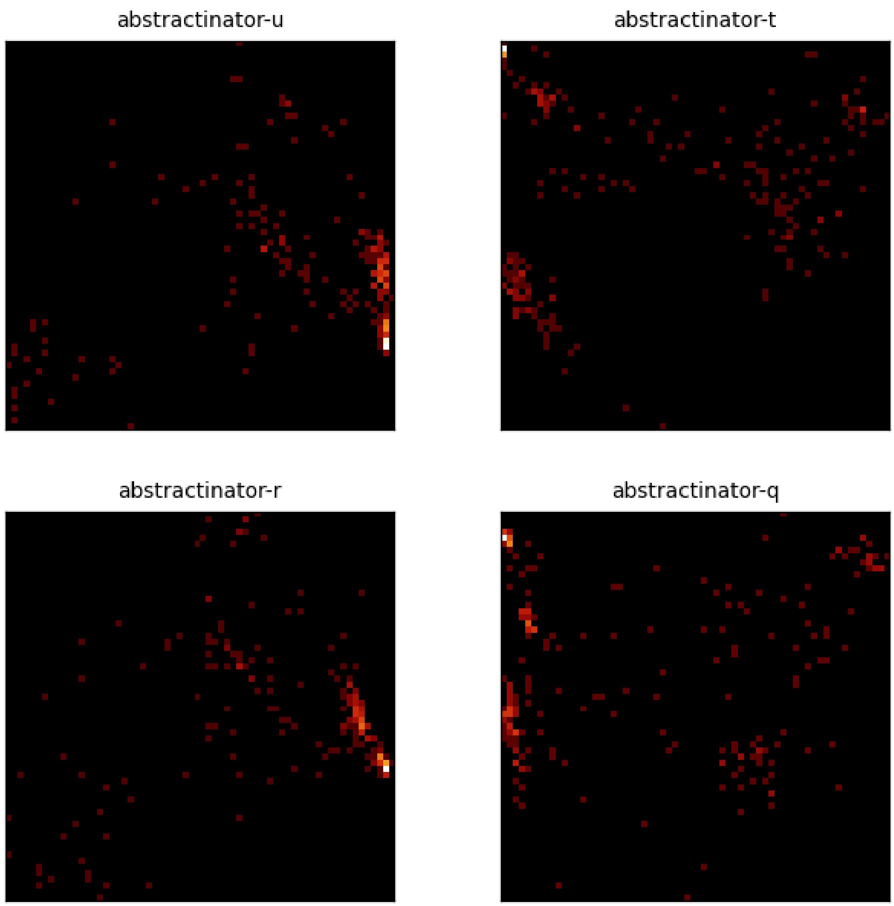



**Figure 12:** Center PCA density for the 4 most recent crystal-head runs (abstractinator-u, abstractinator-t, abstractinator-r, abstractinator-q). Each panel is the top-2 PCA projection of a head's vocabulary centers, binned to a 64×64 density grid (log color) - the same snapshot the dashboard renders live. Different runs settle into structurally different geometries, not noise.

ating point: the field learns which rhythm matters, the delta learns when and how much to deviate, and no per-experiment hyperparameter mediates the two. If the scaling conjecture holds even partially, then on problems with strong periodic or compositional structure, the right move is to change the basis rather than to add parameters, and the linear and logarithmic regimes should be distinguishable by measuring the slope of coverage against resources. Both are testable in the framework.

### 8.8 Why architecture is not neutral ground

A standing objection deserves a direct answer, because the framework's whole premise stands or falls on it. If universal approximation [27] guarantees that a sufficiently wide network of *any* family can represent any function we care about, then any Praxis experiment - however intricate - is by hypothesis emulable by some other network, and one is entitled to ask where the value of the architectural complexity could possibly live. The answer is that the objection conflates three distinct equivalences, and universal approximation establishes only the first of them.

Representational equivalence says the function exists in the closure of each family; it is an existence statement at unbounded width. Learnability equivalence would say stochastic gradient descent, from a random initialization, actually *reaches* that function in feasible compute and data. Statistical equivalence would say two families generalize equally at a fixed sample budget. Universal approximation asserts the first and is silent on the other two by construction - it quantifies existentially over parameters, not over what an optimizer finds or what a finite sample supports. The value of an architecture lives entirely in the second and third layers, which is exactly where the existence theorem says nothing.

The foundational mathematics here cuts *for* the framework, not against it. Depth-separation results (Proposition 2) exhibit functions whose representation cost differs exponentially between architectures that are equal as sets of representable functions: the family is the same in the limit, the resource at fixed accuracy is separated by $2^{\Omega(k)}$. "Architecture is just a constant factor" is therefore not a conservative default but a claim already contradicted by theory; the burden of proof runs the other way. The honest residue is that representation cost is not yet learnability, and proving that a given pair of comparable-capacity architectures have *separated error boundaries under training* - the empirical edge of the conjecture in Figure 10 - requires controlled ablations we have not yet run. We state the hypothesis before we have closed it, and mark the seam plainly: the separation of representable cost is established mathematics; the separation of *learned* error across configurations of equal budget is the measurable prediction the framework exists to test, not a result it already owns.

**Proposition 2.** *Let $\mathcal{F}_A$ denote the set of functions a family of networks with architecture $A$ can approximate to arbitrary accuracy given unbounded width.*

*Universal approximation [27] asserts $\mathcal{F}_A = \mathcal{F}_B$ for any two such families: as* sets of representable functions *they are equal. This equality does not imply equality of* cost *- the width, depth, or sample budget at which a target is reached - and there exist target functions for which the cost gap between two architectures is exponential.*

*Proof.* The first claim is immediate: universal approximation quantifies existentially over parameters at unbounded width, so it constrains only which functions lie in the closure of each family, not the resources at which they are reached. Equality of sets is compatible with any cost profile whatsoever.

The separation is witnessed, not asserted. Telgarsky [54] exhibits functions on $[0, 1]$ that a network of depth $\Theta(k^2)$ and constant width represents exactly, but that any network of depth $O(k)$ requires $2^{\Omega(k)}$ width to approximate. Eldan and Shamir [13] exhibit a function expressible by a depth-three network of polynomial width that any depth-two network must use exponential width to approximate. In both, $\mathcal{F}_A = \mathcal{F}_B$ holds in the unbounded limit while the resource at fixed accuracy is separated exponentially. Hence representational equivalence is the wrong invariant for comparing architectures of bounded size; the relevant quantity is cost, and cost is architecture- dependent by these results. The further descent - from representation cost to what stochastic gradient descent actually *reaches* from random initialization, and to what *generalizes* at a fixed sample budget - is the inductive-bias question, on which approximation theory is silent by construction; the lottery-ticket phenomenon [16] is one empirical marker that two equal-capacity families do not induce the same reachable solutions. We import these as the foundation the framework's conjecture stands on, not as results re-derived here. □

## 8.9 Neither halting nor blocking

A recurring objection holds that a neural network cannot be Turing-complete, and the halting problem is offered as the reason. The invocation is a category error. Turing-completeness is the capacity to express any computation, including ones that never halt; deciding in advance whether an arbitrary program halts is a separate question, and an undecidable one for Turing machines themselves. The theory in fact runs the other way - recurrent networks with unbounded precision and unbounded time are already Turing-complete [46]. What bounds a real model is not undecidability but finite precision and a finite number of steps, exactly the resources a fixed-depth forward pass exhausts and exactly the ones recurrence and delay restore.

Praxis leans into that. With the swarm online, computation runs in two directions at once. Forward, the recurrent loop propels generation and is never required to emit a stop: the halting gate is a convergence check, not an exit, so generation is a continuous autonomous stream rather than a program that terminates. Sideways, a pool of remote experts votes - stochastically sampled, mixed by an expert vote in the style of CALM's patch vote [44] - and trains without blocking, so a slow peer never stalls the others; its update simply lags and lands on a later pass. Computation is lagged as much as propelled, and the model commits to neither a halt nor a barrier. It approximates both by doing neither.

This relocates the question. It is no longer whether the network halts but what the pool expresses, over how much time, at what precision - the resources, not the stop bit. The butterfly is the right image: one expert, then two, then four, each a wing whose small vote the mixture amplifies or cancels, the whole field sensitive to where the late votes fall. The interior turns of the recurrent loop (Section 5) already carry the decision; the last is only the budget running out. Power here is bought with peers and patience, not with a proof that the machine knows when to stop.

The same shape is built into the interface, where it can be used rather than only argued. A network is Turing-complete only with constraints applied - finite precision, a finite step budget, and some structure on what it may do - and Praxis makes one such constraint visible as the Gymnasium, a read-eval-print loop turned from a sequence into a choice. A classical REPL runs its four stages in series - read, then evaluate, then print, then loop - each blocking on the one before. Praxis treats them not as stages but as actions and activates one at a time: *Read* the knowledge base, *Evaluate* a conversation, *Print* a question of its own, with the recurrent loop the implicit fourth that carries the stream onward - an XOR over the handful where the conventional loop is their dense, ordered product. Sparse, mutually exclusive action selection is the halting-and-blocking argument in miniature: a loop that neither marches a fixed sequence to a terminating print nor blocks at each stage, but picks a single move and continues. Read and Evaluate are live; Print is the planned next button on the row, and the constraint is the point - the machine is general because of what it is forbidden to do all at once, not in spite of it.

A living document can also look at itself. Praxis is built in the open, commit by commit, and that history is just another sequence - a stream of byte-level diffs over time. Figure 13 turns the framework's own recency kernel on its repository: per-subsystem churn across the project's history, with the distant past faded in proportion to its distance from the present, the way a decay bias weights earlier positions in a sequence. It is not an argument, only a mirror - the same machinery that reads a corpus, reading the record of its own construction. The reading that any finite byte sequence is a position to be classified the same way, the repository included, is left as a conjecture to test rather than a claim to lean on (see the project notes).

### 8.10 Standing conjectures

We gather the paper's falsifiable conjectures in one place; each is discussed where it arises. They are bets about reachability and cost, not theorems, and each names what would refute it.

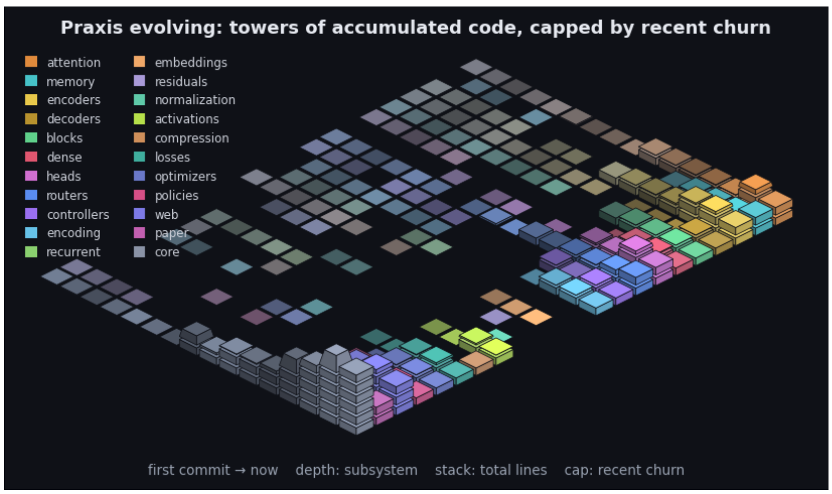



**Figure 13:** Praxis's git history as an isometric terrain over the last 2603 commits. Each cell is one subsystem in one time window: a stack of base blocks for its accumulated size (total lines, recency-weighted) topped by a vivid cap for that window's line churn - the standing codebase carrying the recent activity. Both are weighted by recency (the kernel the model applies to a sequence, turned on the repository): recent windows stack tall with bright caps while history thins toward the always-present prior - the low valley we roll into. Color fades from each subsystem's hue toward a neutral prior into the past, banded in phased strata over a non-linear timescale. It is the bias/variance geometry read along the frequency axis (see the project notes): loud recent corrections over the quiet, ever-present base. By recency-weighted churn the current center of gravity is core, web, paper. The framework charts the hand that builds it.

- **Linear consensus, logarithmic interference.** Learning a corpus as a consensus mean costs resources linearly in the patterns expressed, while composing it by harmonic interference costs only logarithmically, so the coverage-versus-resources slope a basis buys is measurable and basis-dependent. The representational half is no longer conjectural - the amplitude-to-field map is an isometry, so a $K$-dimensional resolution grid carries to exponentially many separated fields - which leaves two things to measure: whether gradient descent reaches those configurations, and whether the spectrum stays spread enough to hold them, since capacity saturates in $K$ under a fixed envelope with bounded amplitudes. [28]

- **Bias and variance as orthogonal axes.** Bias and variance - the projections of the harmonic latent onto its frozen, time-invariant modes and its time-varying modes - occupy orthogonal subspaces whose loss cross-curvature vanishes, so under training both fall together toward the origin rather than trading off along a frontier; the claim is falsified the moment the measured (bias, variance) trajectory bends into a Pareto frontier, or the cross-curvature between the two mode bands stays bounded away from zero. Orthogonality alone is not sufficient: by the interference-capacity proposition the variance axis carries configurations only while the input-conditional spectrum stays spread, so a delta whose amplitudes concentrate alongside the static baseline - or decay with the $1/f^{\alpha}$ envelope instead of resisting it - leaves the axes orthogonal and the second one empty, which refutes the useful form of the claim.

- **Always grokking.** Because the grokked solution is itself a harmonic representation, a model with frozen phases begins post-grok and never undergoes the delayed memorize-then-generalize transition over training; the transition instead recurs on the input axis - the symbol set is fixed, but the belief state is perpetually re-fitting, hopping between those fixed symbol clusters across compute steps - and the claim is falsified the moment that per-input path is measured to be a smooth drift rather than discrete cluster-to-cluster hops, or the symbol set is found to keep growing rather than being recombined. [39, 43]

- **Development reverts to the basis.** If the contortion of the hidden state across recurrent-depth steps is a wind-up that releases back to the frozen harmonic basis - development expressed and then reverted, in the manner of cumulative selection - then the per-step state trajectory should be stable, close to the standing harmonic shape, punctuated by sparse discrete jumps in magnitude rather than a continuous drift, and the claim is falsified the moment that trajectory is measured to drift smoothly and monotonically away from the basis without returning, or to change at every step rather than in rare punctuated events. [11, 14]

- **Information density at the rim.** Information density - the share of the representation's distinguishing content carried at each position - is uniform by assumption in the consensus regime but pushed to the window's extremes by a harmonic latent, a slow hum anchored at the head and an input-conditional chirp at the tip, so hidden-state deviation must grow with recurrent depth under a monotone positional gradient - early positions settling toward a fixed point while the tip stays live - wherever a recurrence is harmonized; the claim is falsified the moment the per-position, per-step deviation profile - read in norm and in symbol occupancy alike, since a silent bit flip can change the geometry without moving the norm - is measured flat in position, or fails to steepen across depth. [61, 50]

- **Reachable, not merely representable.** Across architectures of comparable parameter count and compute there exist task and data regimes where generalization error is separated, because the prior sets which solutions stochastic gradient descent can reach, not only which functions exist in the closure. [54, 16]
- **Lottery engineering.** A geometric prior makes good sparse subnetworks reachable by stochastic gradient descent directly rather than only by train-then-prune, and a rotating population of such subnetworks (mixture-of-widths) approaches full-width performance at lower average active width. [16]

### 8.11 Limitations

We are deliberate about what is and is not established. The bias-variance decoupling is grounded in a running implementation and its diagnostics (concentration, smoothness, delta norm, the spectrum heatmap), but “orthogonal axes” is an interpretation of those diagnostics, not a theorem. The scaling conjecture of Figure 10 is a conjecture: the figure draws an asserted relationship, not measured curves, and the honest next step is to measure the slope on tasks of varying combinatorial structure and report where harmonic representations fail to buy it. The observer-relative view of tractability (Section 5) is a philosophical position, and only its weak form - the scaling slope a given basis buys on a given problem family - is measurable; we make no claim about the formal worst-case $P$ versus $NP$ question, which abstracts the observer away by design. The connection to biological attention remains speculative, and the prismatic asymmetry needs to be shown in silico now that the hand-measured figure is gone. The observer effect of Section 7 is, in its quantum form, an analogy we do not cash; only its weak form is a claim - that impure measurement (state-mutating reads, measurement-induced rounding on the exponential edge, self-ingested execution timestamps) perturbs a non-local harmonic field measurably more than an independent-feature one, and that coupling the framework’s passive observer pool back into the forward is a two-arm ablation of being-watched. The temporal artifact in particular is a claim about data provenance, not physics: launch rates and license stamps enter the model as tokens, so “modeling its own time” means modeling those tokens, and we make no claim that the field encodes execution timing beyond what is literally written into its corpus. Those are diffs between a watched and an unwatched run, not statements about physics. The coupled-observer extension (Section 8.2) - that two phase-locked observers amplify perturbations of the frame they share, of which *feeling watched* is the folk instance - is conjecture in the same spirit: its only measurable form is the synchrony-stratified design that separates a resonance gain on ordinary sensory cues from a non-sensory channel, and we side explicitly with the former, against the latter, pending that experiment. None of these are reasons not to state the ideas; they are the boundary between what we have built and what we are conjecturing, drawn so the reader can attack the right target.

### 8.12 A direction: the moving point

One reading the geometry keeps inviting is worth stating as we close, marked as a direction rather than a result. The case for the third state was that two points fix only a line and three fix a plane (Lemma 4), and that a phase can turn only in a plane - so the least representation that can hold an oriented shape is two-dimensional, not one. But that plane is only where a phase *sits*. A phase also moves: it carries a velocity and a direction, and a position taken together with its motion is a phase space, twice the dimension of the configuration it lives on. The line becomes a plane, and the plane the harmonic field was built to turn in becomes the four-dimensional space in which that turning is itself a coordinate. The representation we have described is therefore never a point; it is a moving point, and the climb - from a position, to a line, to the plane where phase lives, to the phase space where phase moves - is the same ascent at every scale. We make no claim that the network parameterizes that space explicitly. We note only that the geometry, followed honestly, does not stop at the plane: velocity and direction, not position alone, are what a harmonic representation is finally about.

**Lemma 4.** *Two distinct points determine a line; three non-collinear points determine a unique plane. A phase - a point on the circle $S^1$ - is realizable only in dimension at least two, so three points are the minimal affine data that orient the plane a harmonic component turns in.*

*Proof.* Through $p_0, p_1, p_2$ the affine span $\{\, p_0 + s(p_1 - p_0) + t(p_2 - p_0) \,\}$ is a plane exactly when $p_1 - p_0$ and $p_2 - p_0$ are linearly independent - that is, when the three points are non-collinear; two points give a single direction and hence only a line. Since $S^1$ embeds in $\mathbb{R}^2$ but in no line, a continuous phase requires that plane, and three points are the fewest that fix it.

# 9 Conclusion

We began with scale and ended with geometry. The patterns a model can learn are set by the shape of the space its weights may occupy, and a harmonic latent space reshapes the oldest tradeoff in the field: bias becomes a static, population-average spectrum, variance becomes an input-conditional modulation, and the paper's structural claim is that they live on orthogonal axes and can therefore be lowered together. From that decoupling follows a conjecture stated plainly -

that learning the mean costs linearly while composing by interference costs logarithmically - and a framework built to test it rather than to argue it. And from that, in turn, follows the harder claim: that tractability is not a fixed property of a problem but a relation between a problem and the observer solving it. The same instance is exponential to one architecture and nearly free to another, so $P$ and $NP$ are better read as a spectrum that the choice of representation indexes than as a wall every solver meets in the same place. The speculative figure is gone; the conjectures that replace it are falsifiable. What remains is praxis: the answer manifests through direct experimentation, and what you choose to attend to determines what we find next.

# A Current Experiment

*This appendix is generated.* The figure is built from training logs by the framework's own paper build; the paper itself is a template that build populates. The chart leads with the most recently launched experiment that has logged validation data, and fixes the y-metric to that run's family - here Bits per Byte, as of the last export (2026-09-10). Only experiments that report the *same* metric are drawn alongside it; the rest are listed as omitted below. This is the smallest living slice of the document: one metric, the current experiment against its compatible predecessors.

*Note:*

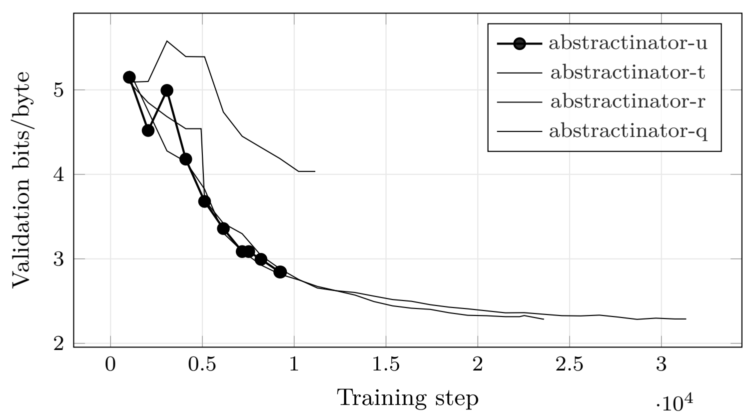



**Figure 14:** Bits per Byte across the 4 most recent experiments with comparable validation curves; the current experiment (abstractinator-u) is drawn in bold. Bold marks *which run this build is reporting on*, not which run scores best. The curves end wherever each run stopped, so they are at different points along their own training, and their endpoints are not a ranking. Generated from run logs - metrics that are not comparable across families (token loss vs. byte bits-per-byte vs. codec/CALM BrierLM) are excluded by construction, so the curves on this axis mean the same thing.

The current experiment, abstractinator-u (`b45965f7b`), reached Bits per Byte 2.844 by step 9260 - the value at the point this build was taken, not a converged result, and not a claim against the other curves, which stopped at step counts of their own. The structural figures elsewhere in this paper are read from this same run, so they are likewise a snapshot of a run in progress rather than of a settled model; that is what makes them a live reading and also what limits what they can be asked to prove. Recent experiments whose families report a different validation metric are omitted rather than plotted on an incomparable axis: .

| Dataset | Type | abstractinator-u | abstractinator-t | abstractinator-r | abstractinator-q |
|---|---|---|---|---|---|
| praxis | directory | 1 | 1 | 1 | 1 |
| synthetic-tool-calling | synthetic | 1 | 1 | 1 | 1 |
| fineweb-edu-350bt | huggingface | 1 | 1 | 1 | 1 |
| fineweb | huggingface | 1 | 1 | 1 | 1 |
| finepdfs | huggingface | 1 | 1 | 1 | 1 |
| persona-chat | huggingface | 1 | 1 | 1 | 1 |
| wildchat | huggingface | 1 | 1 | 1 | 1 |
| natural-instructions | huggingface | 1 | 1 | 1 | 1 |
| cosmopedia-v2 | huggingface | 1 | 1 | 1 | 1 |
| smoltalk | huggingface | 1 | 1 | 1 | 1 |
| git-history | git_history | 1 | 1 | 1 | 1 |
| platformer | directory | 1 | 1 | 1 | 1 |
| rift | directory | 1 | 1 | 1 | 1 |
| vtx | directory | 1 | 1 | 1 | 1 |
| src.eco | directory | 1 | 1 | 1 | 1 |
| eliza | directory | 1 | 1 | 1 | 1 |
| fold | directory | 1 | 1 | 1 | 1 |
| ode | directory | 1 | 1 | 1 | 1 |
| dimanet | directory | 1 | 1 | 1 | 1 |
| fvn | directory | 1 | 1 | 1 | 1 |
| ext | directory | 1 | 1 | 1 | 1 |
| ink.university | directory | 1 | 1 | 1 | 1 |
| l.ink | directory | 1 | 1 | 1 | 1 |
| reaper | directory | 1 | 1 | 1 | 1 |
| cv | directory | 1 | 1 | 1 | 1 |

**Table 1:** Training-data mixtures behind the 4 most recent experiments, resolved from each run's recorded configuration: named collections expand to their member datasets, and every explicitly configured data directory is its own sampler. Cells are the configured per-dataset sampling weights (pre-normalization), not token counts; – marks a dataset absent from that run's mixture.

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